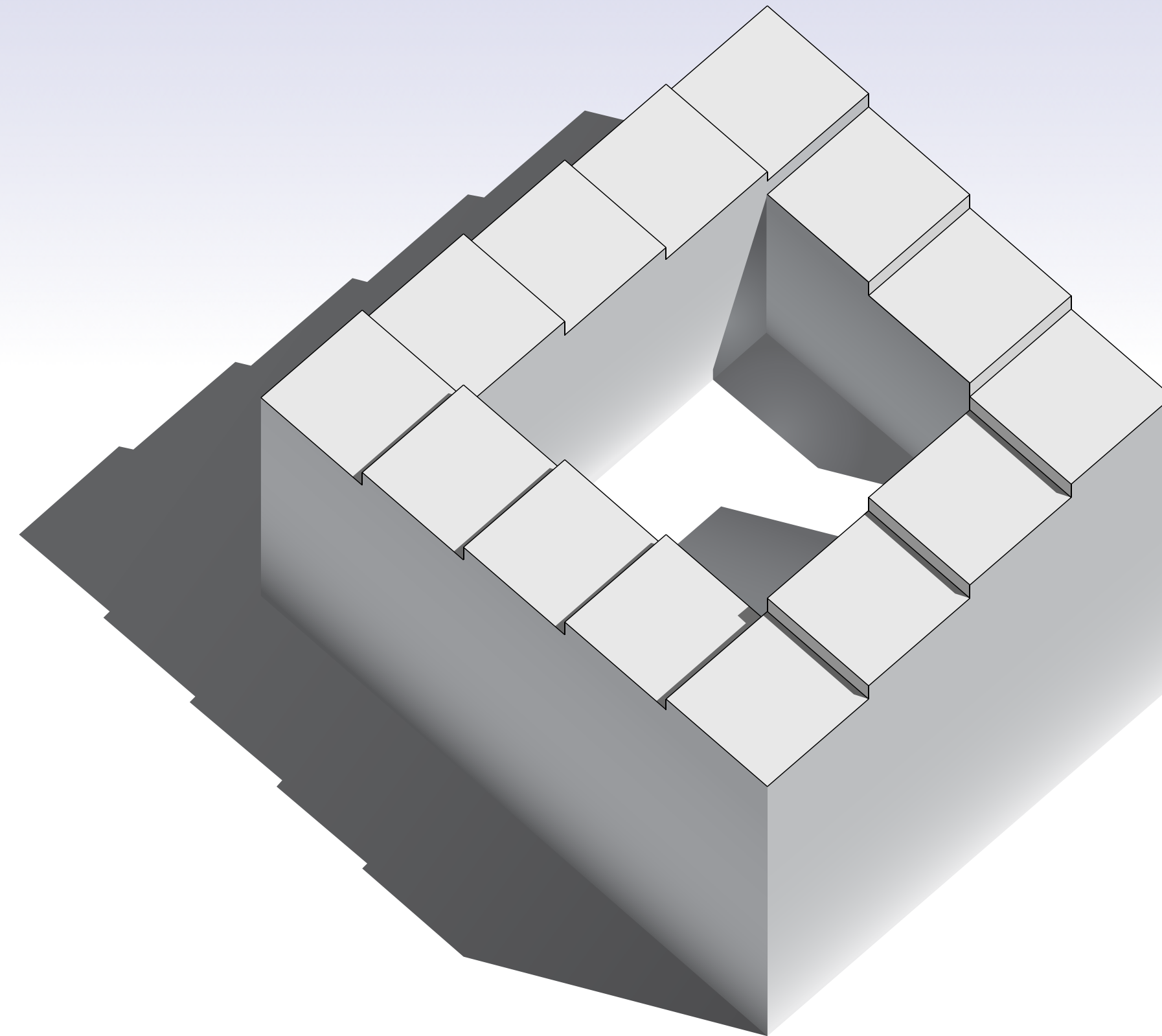
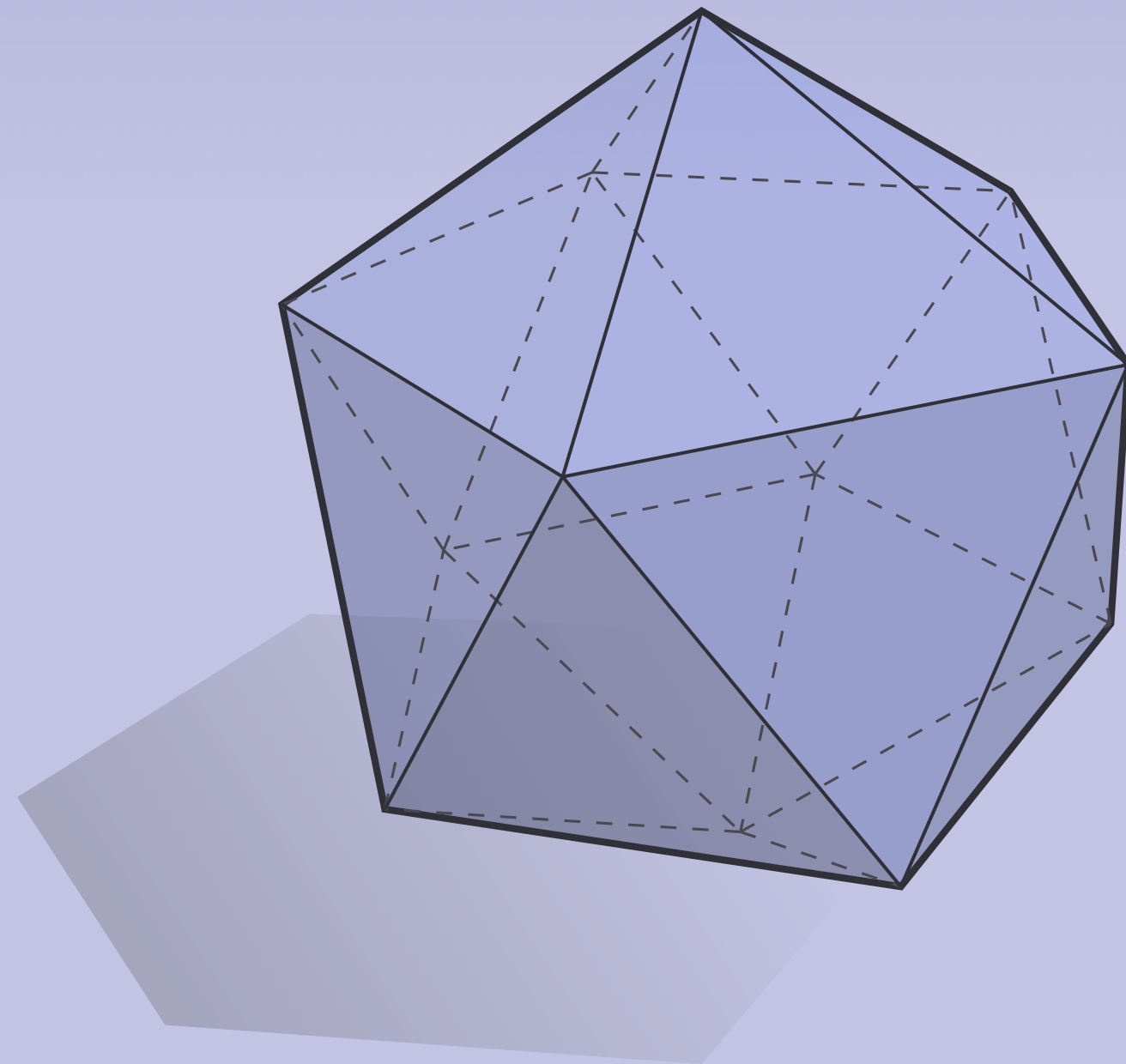




DISCRETE DIFFERENTIAL
GEOMETRY:
AN APPLIED INTRODUCTION
Keenan Crane • CMU 15-458/858

GEODESICS I

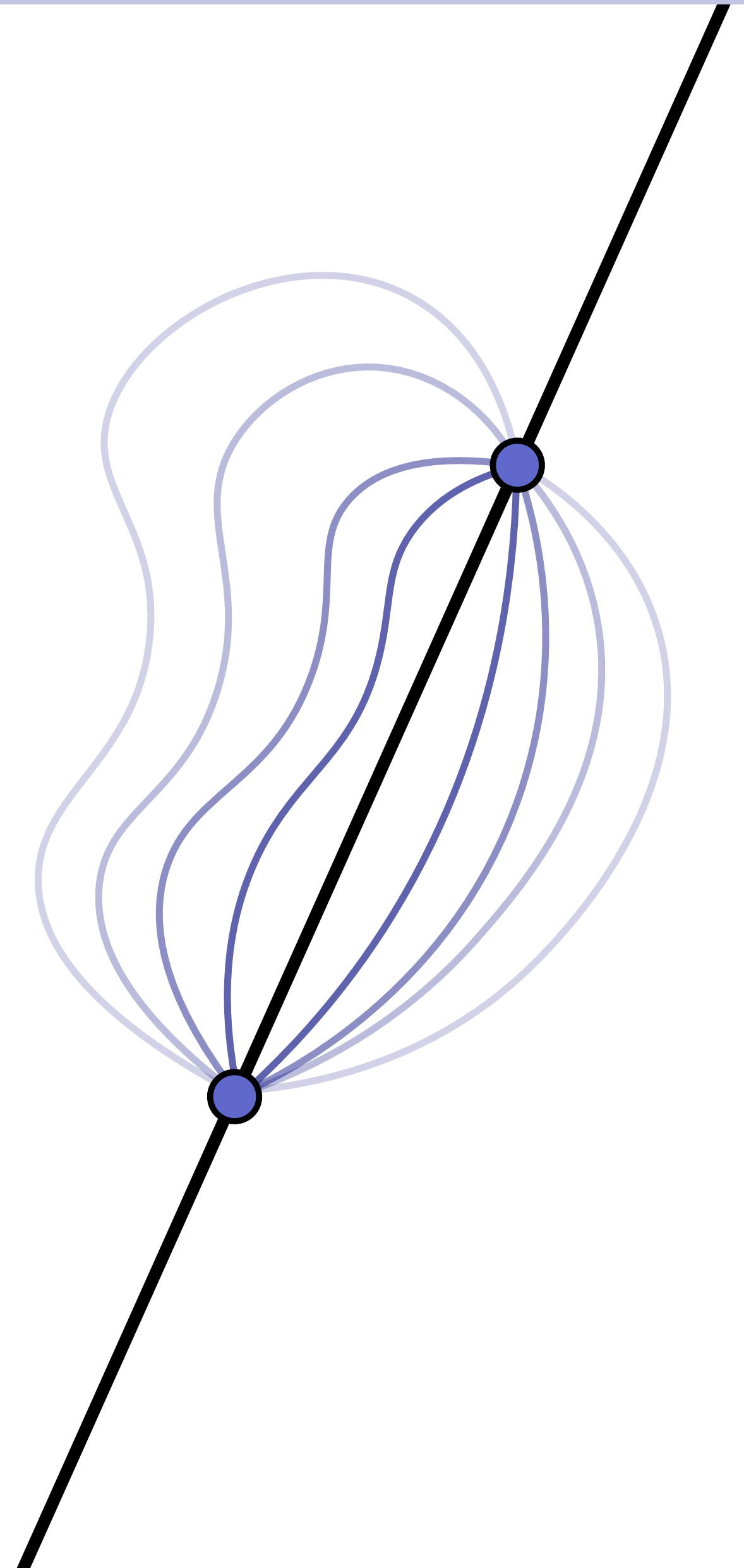


DISCRETE DIFFERENTIAL GEOMETRY: AN APPLIED INTRODUCTION

CMU 15-458/858 • Keenan Crane

Geodesics — Overview

- Generalize the notion of *lines* to curved spaces
- Ordinary lines have two basic features:
 1. straightest — no curvature / acceleration
 2. shortest — (locally) minimize length
- Geodesics share these same *local* properties, but may exhibit different behavior *globally*
- Part of the “origin story” of both classical and differential geometry...



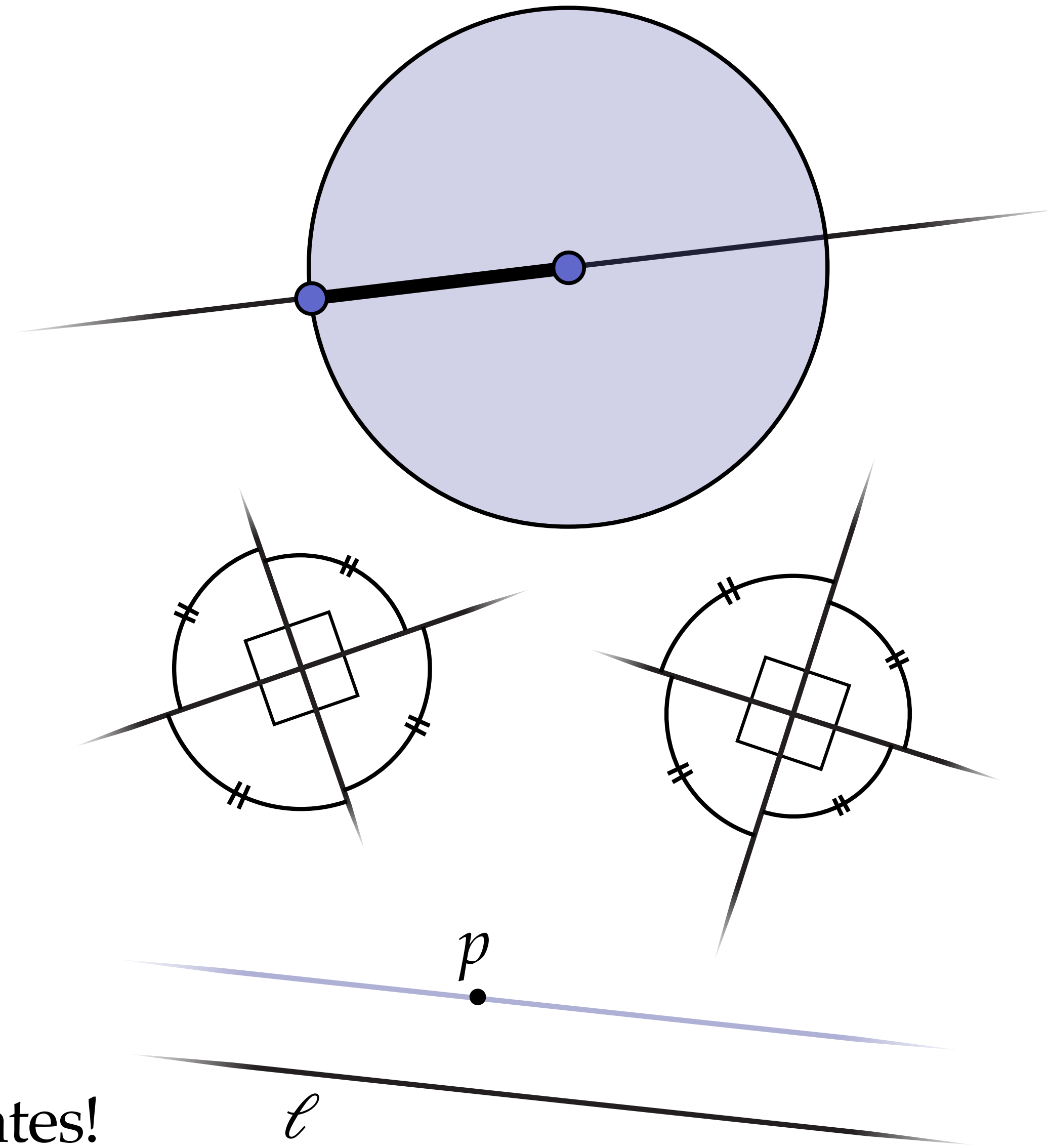
Key idea: geodesic is straightest, (locally) shortest curve

Euclidean Geometry

Euclid (c. 300BC) used five basic “postulates” as a starting point for geometry:

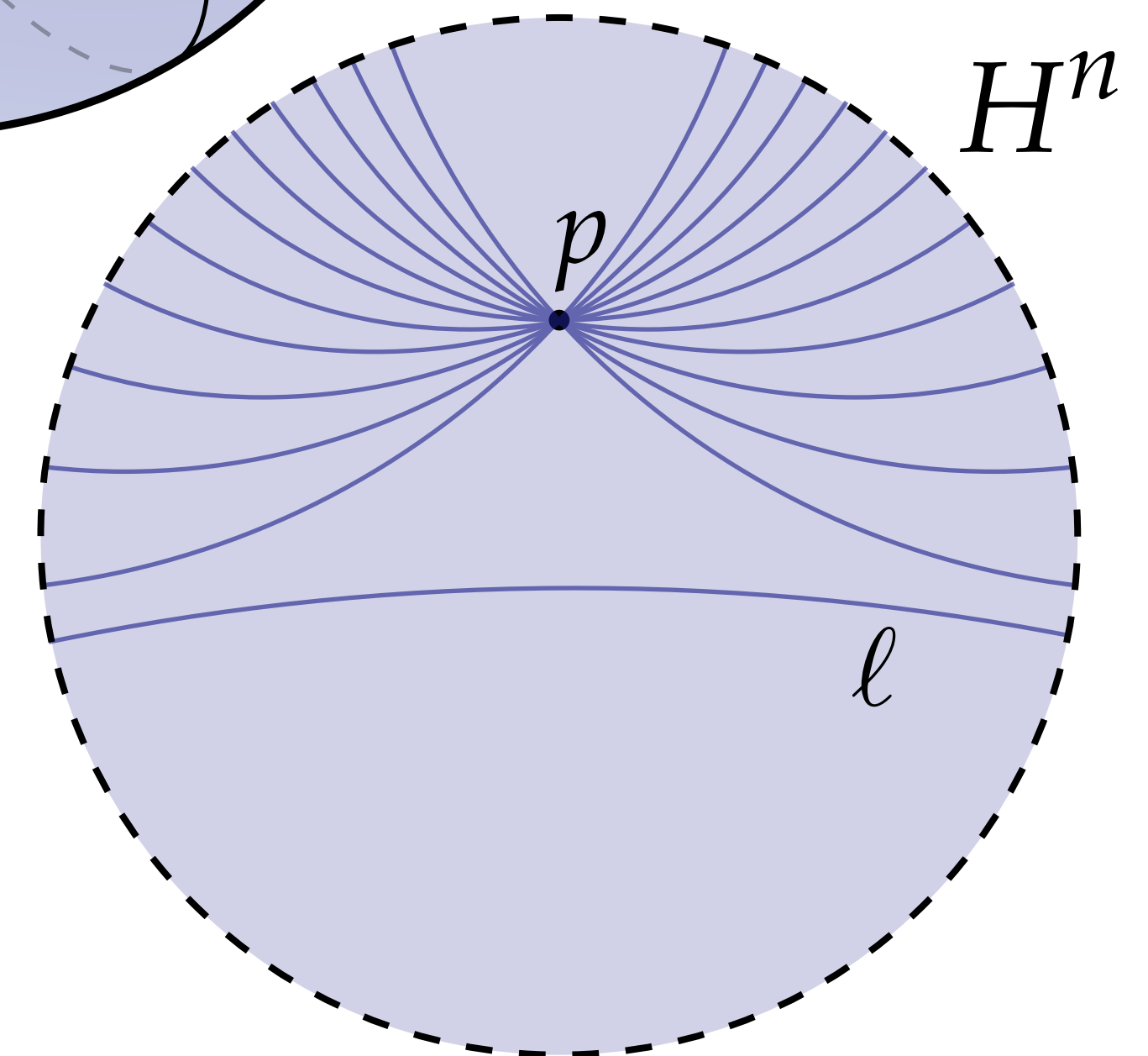
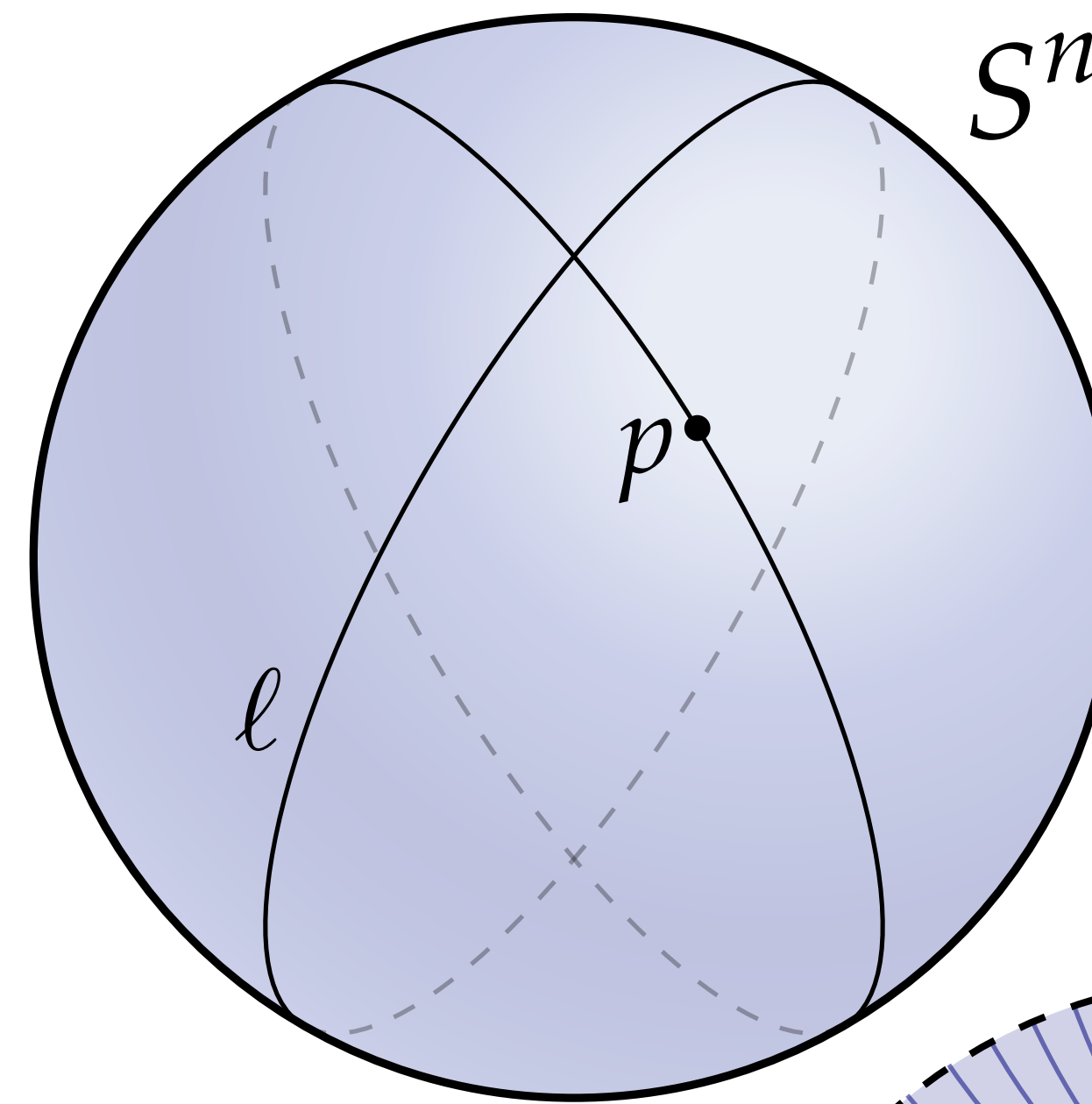
- I. Any two points can be connected by a straight line segment
- II. Any line segment can be extended into a line
- III. For any segment, there's a circle centered at one endpoint, with the segment as a radius
- IV. All right angles are congruent
- V. For any line ℓ and point p not on ℓ , there's a unique line parallel to ℓ passing through p

Idea: *everything* else can be proved from these postulates!

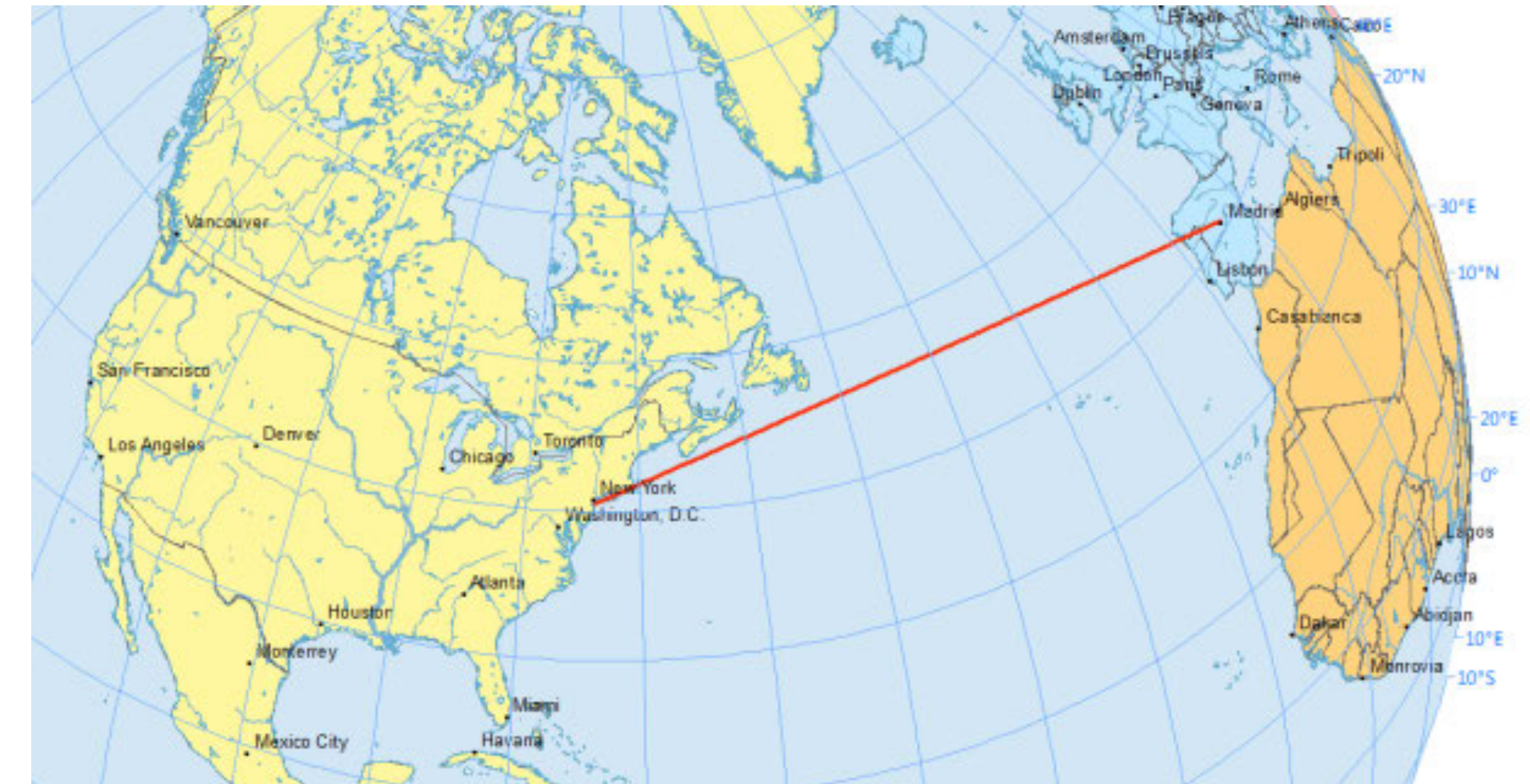
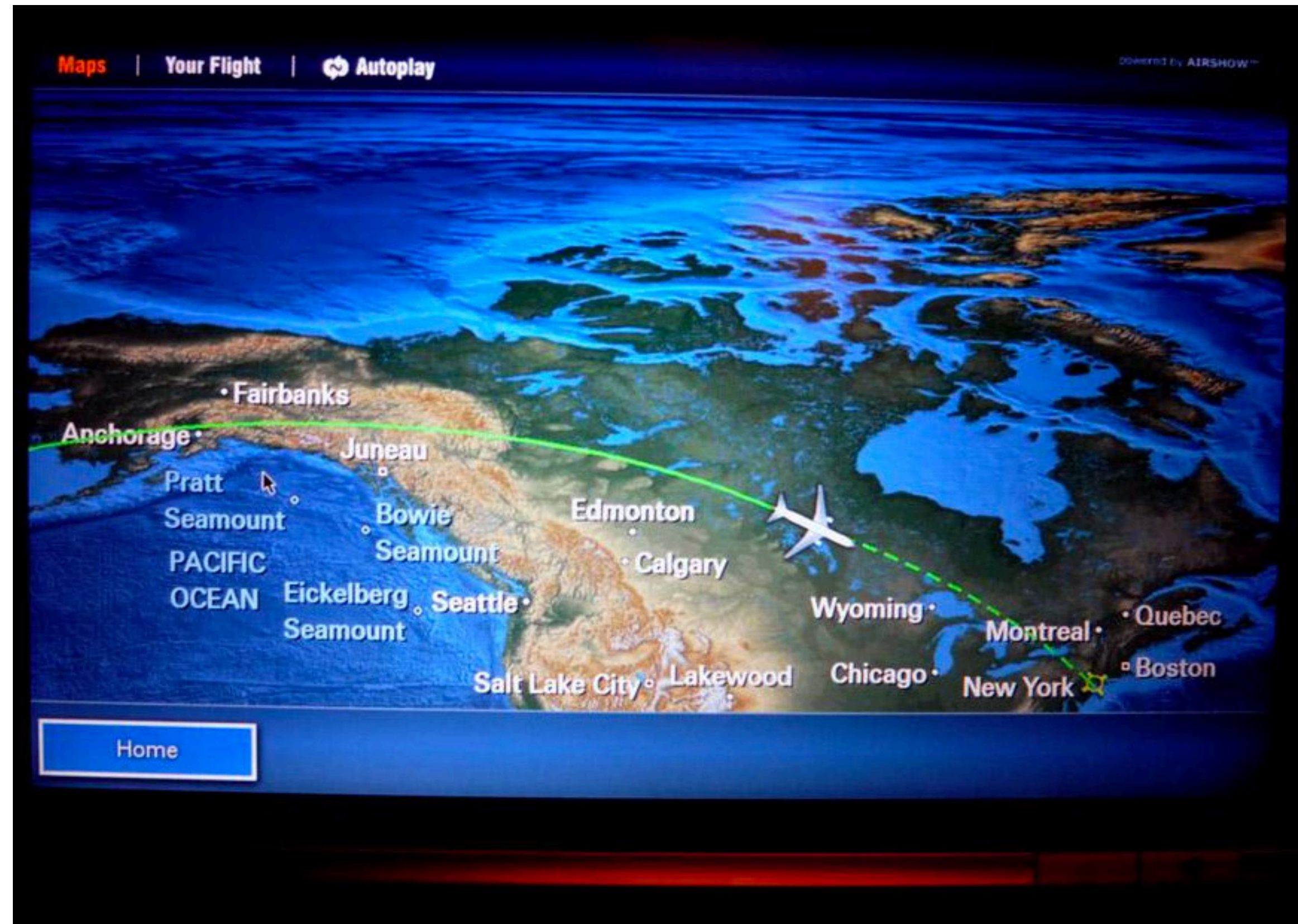


Non-Euclidean Geometry

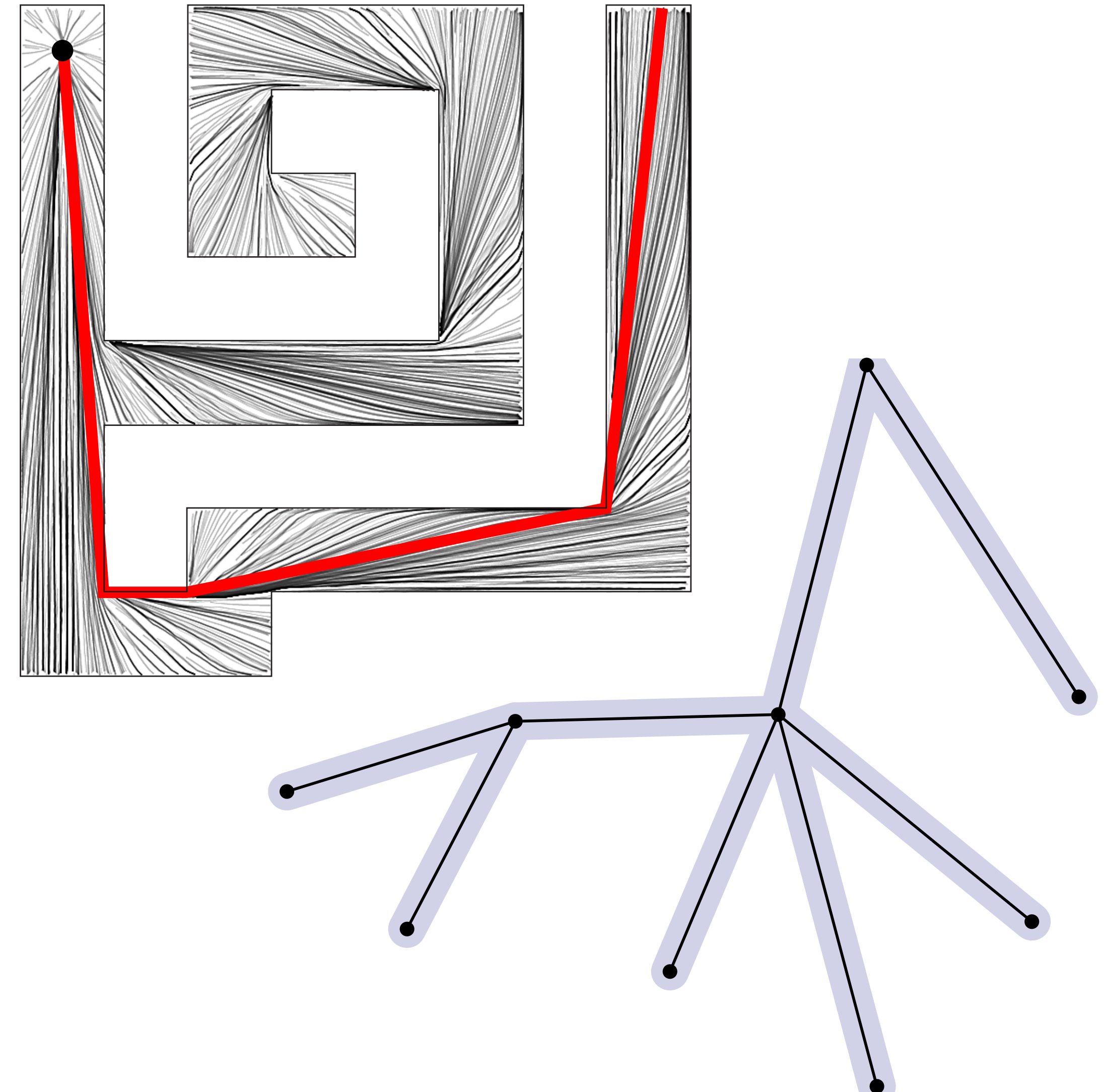
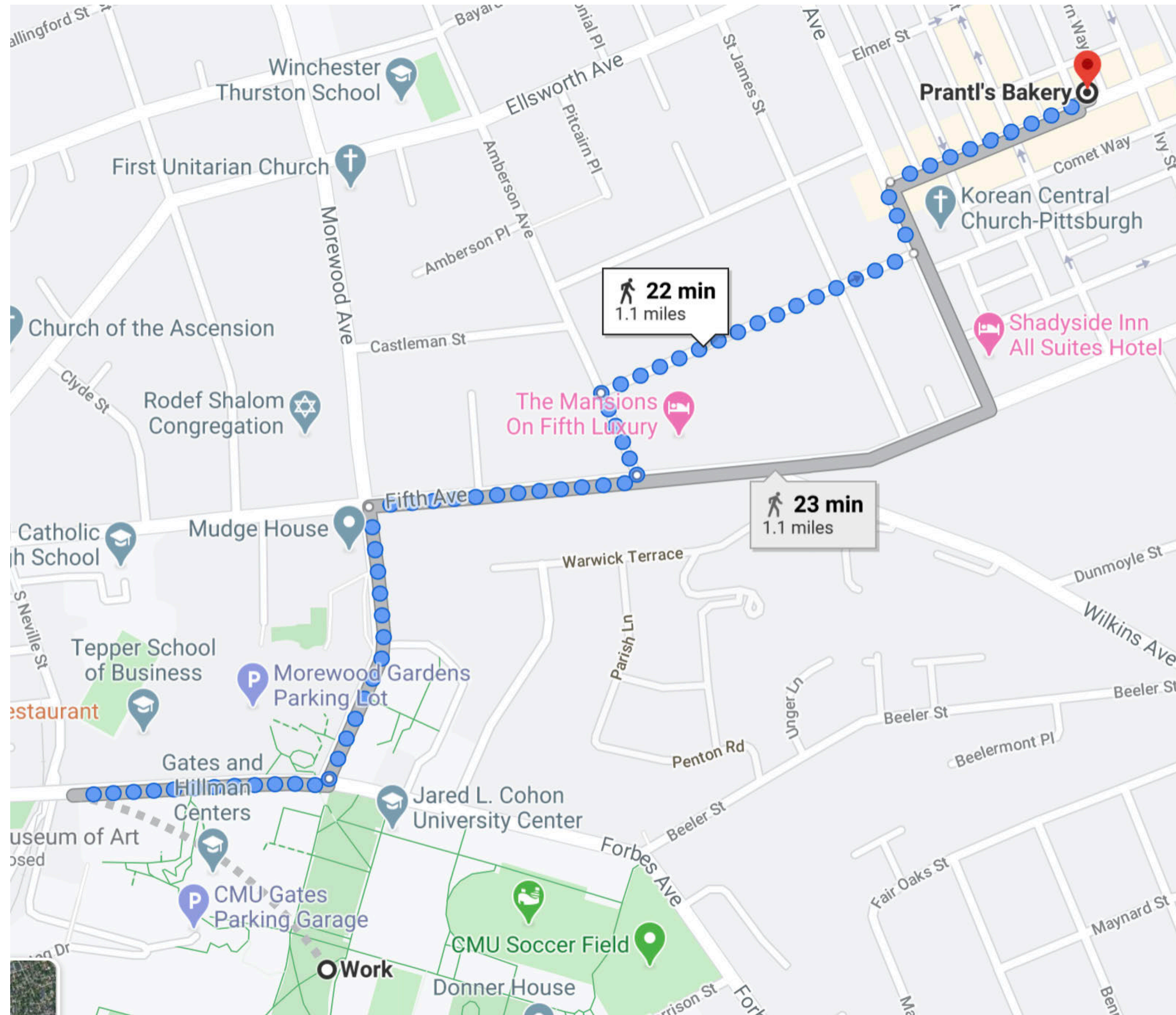
- *Many* attempts to prove parallel postulate from first four. After two thousand years...
- (Lobachevsky, Bolyai, Gauss, ...) Not possible! There are other logically consistent geometries where parallel postulate doesn't hold:
 - **Elliptic:** no parallel lines through a point—all lines intersect
 - **Hyperbolic:** parallel line through point is not unique
- More generally: “lines” or *geodesics* on curved surfaces will behave differently than in the plane
 - Will try to understand this behavior today...



Examples of Geodesics—Great Arcs on the Sphere

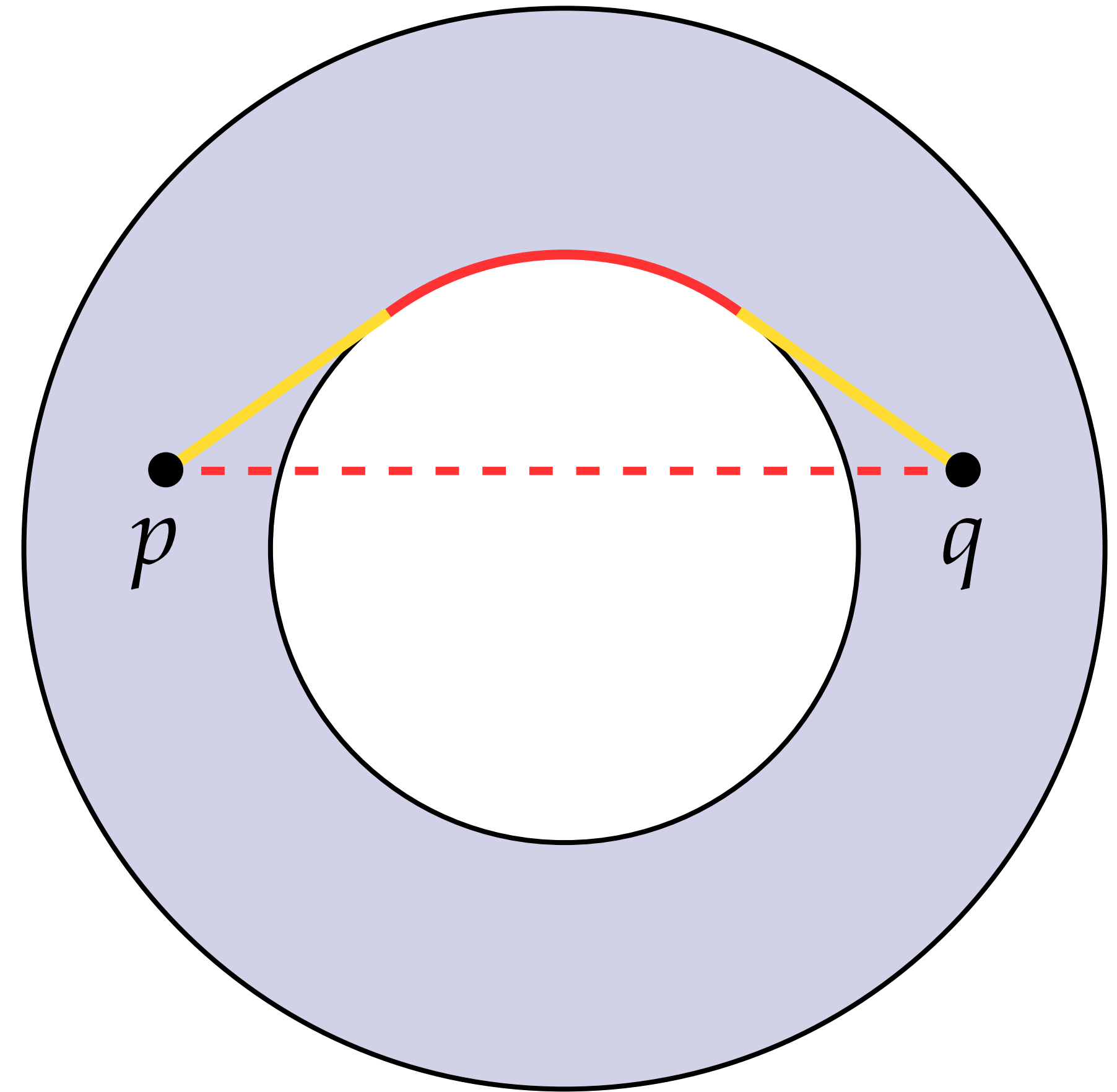


Examples of Geodesics—Shortest Paths in Graphs



Aside: Geodesics on Domains with Boundary

- On domains with boundary, *shortest* path will not always be *straight*
 - can also “hug” pieces of the boundary (curvature will match boundary curvature, acceleration will match boundary normal)
 - on the interior, path will still be both shortest & straightest
- For simplicity, we will mainly consider domains without boundary



Examples of Geodesics—Paths of Light

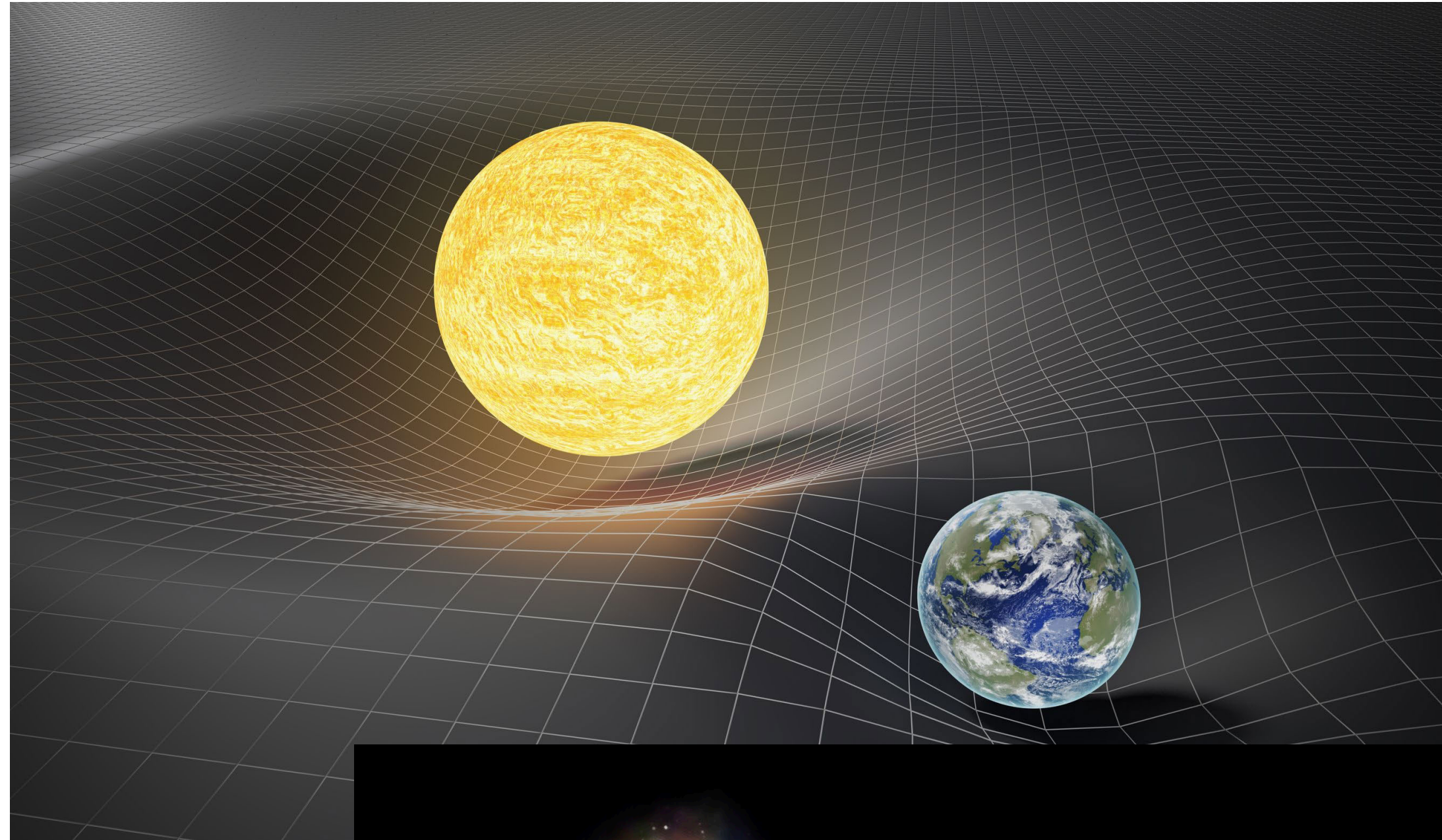


image credit: European Space Agency

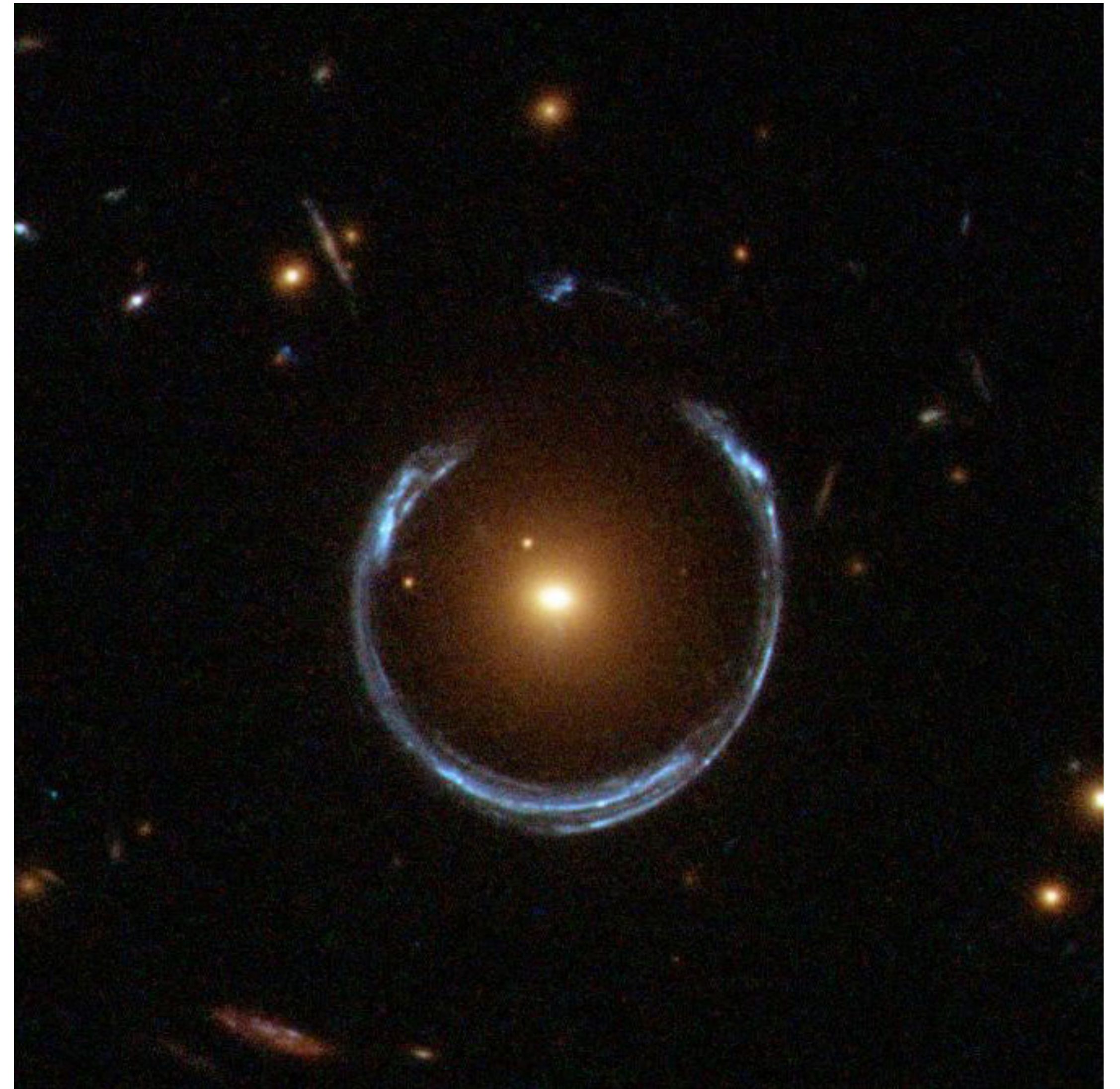
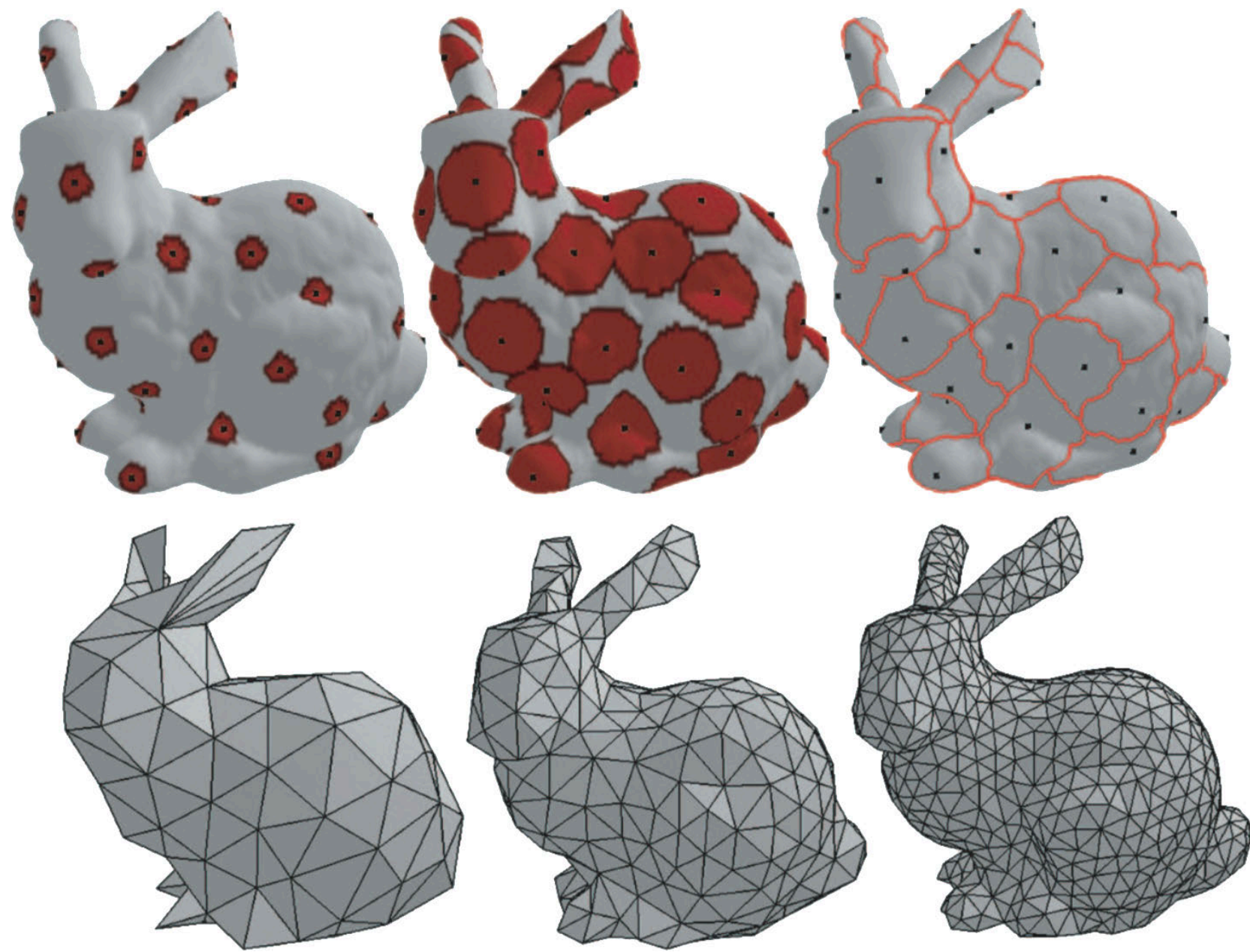


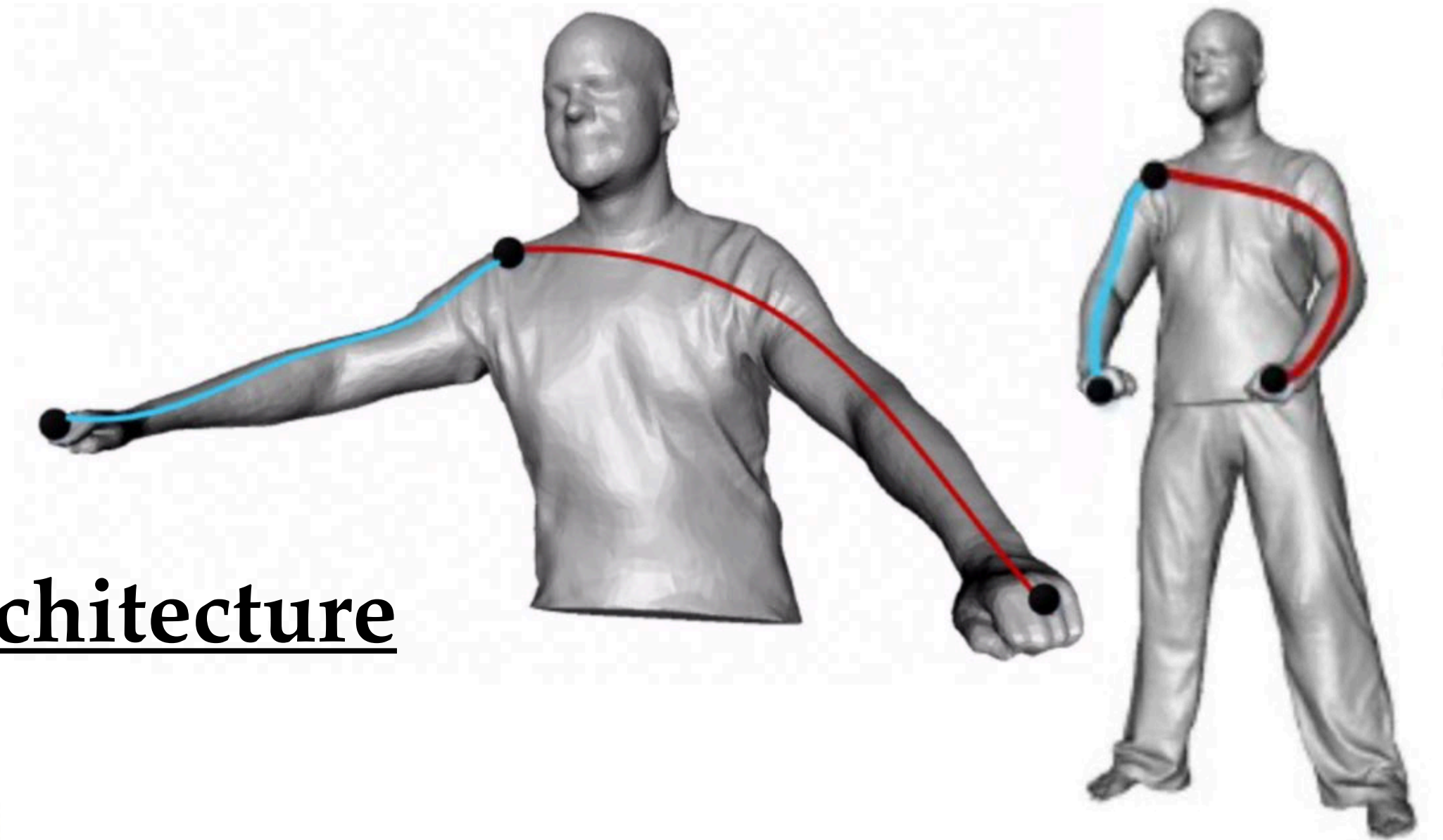
image credit: ESA/Hubble & NASA

Examples of Geodesics — Geometry Processing

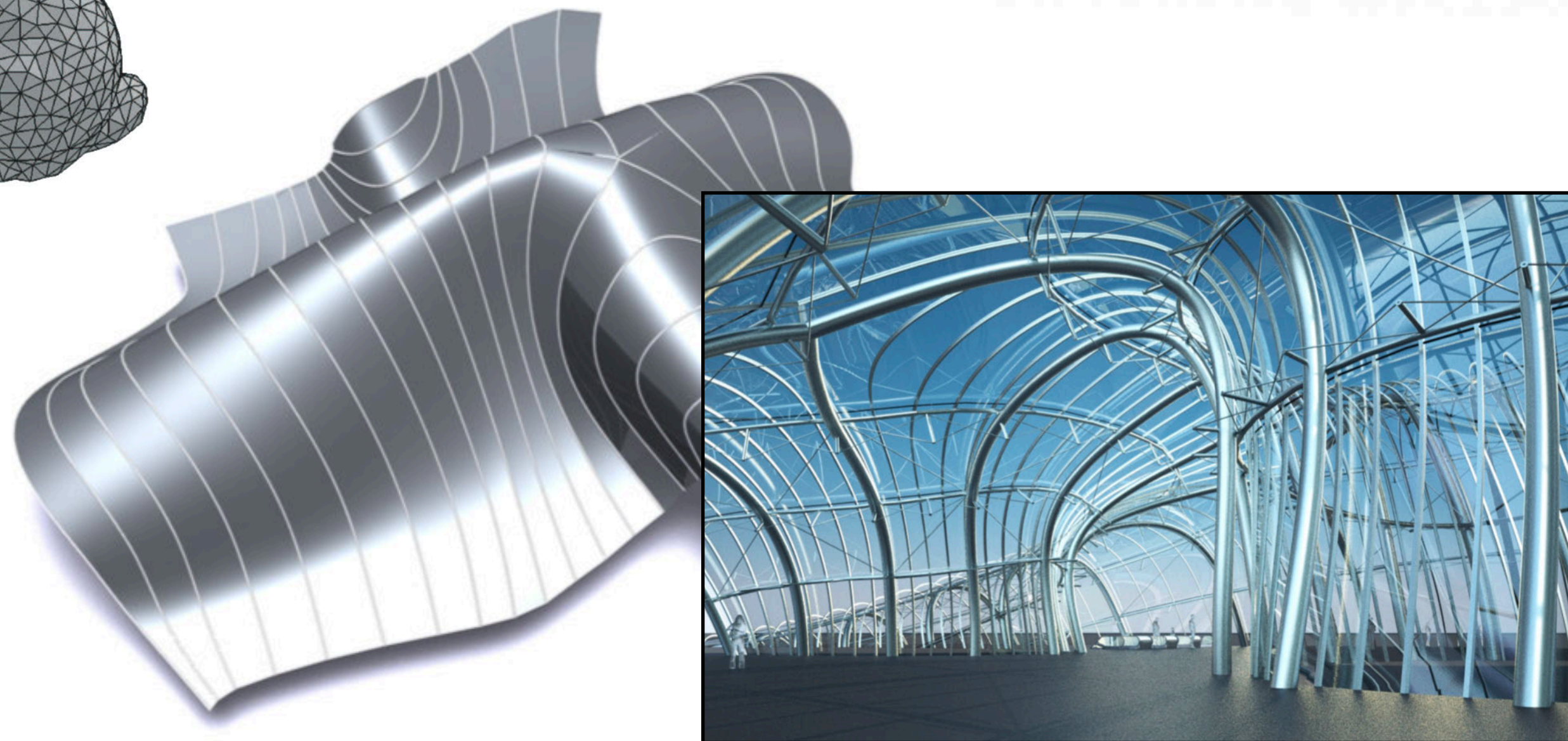
surface remeshing



shape analysis / correspondence



computational architecture



Isometry Invariance of Geodesics

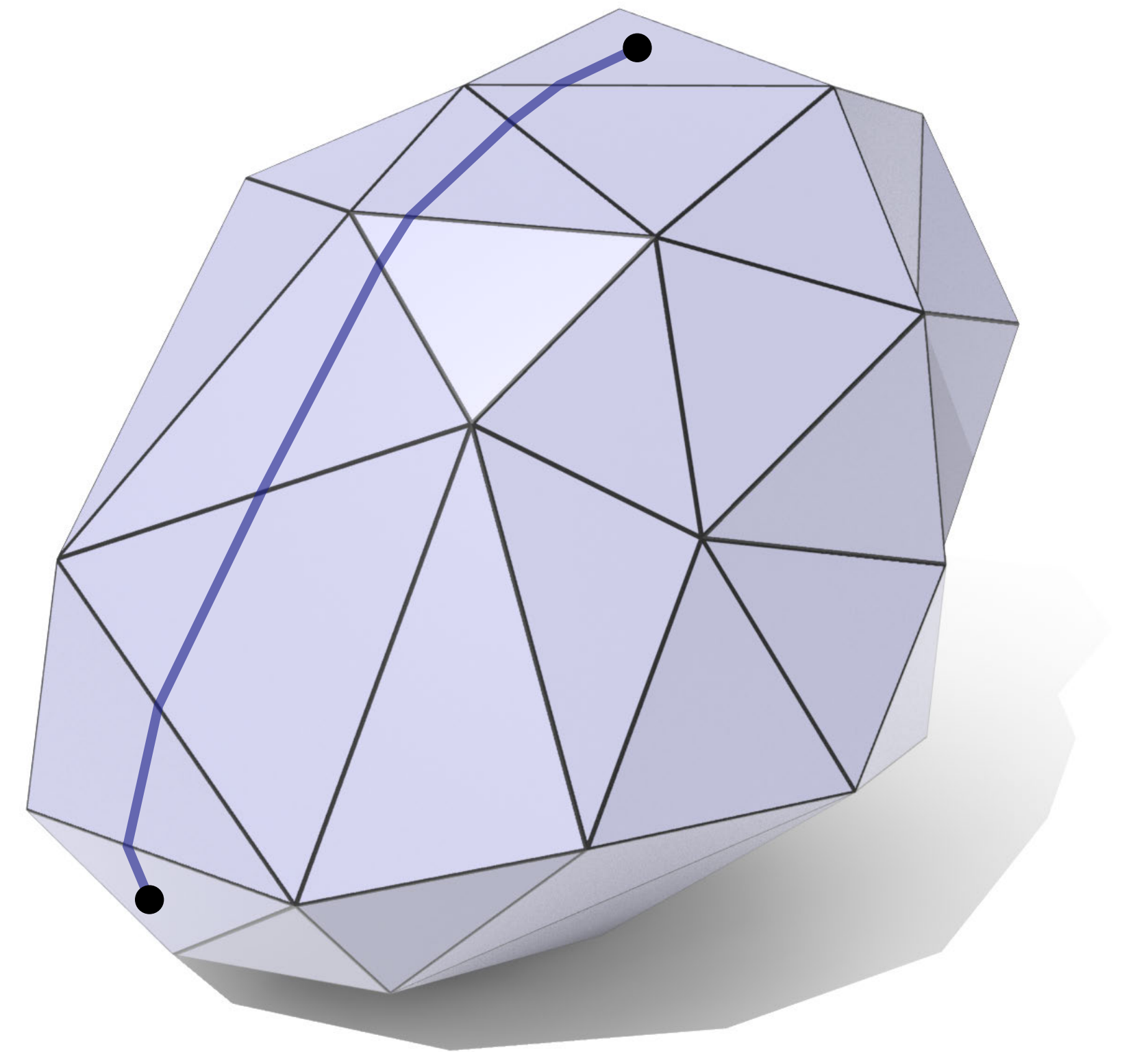
- *Isometries* are special deformations that do not change the intrinsic geometry
 - Formally: preserves the *Riemannian metric* (which measures lengths & angles of tangent vectors)
- For instance, folding up a map doesn't change angle between north and south, or areas of land masses
- Likewise, the shortest path between two cities does not change if we roll up a map



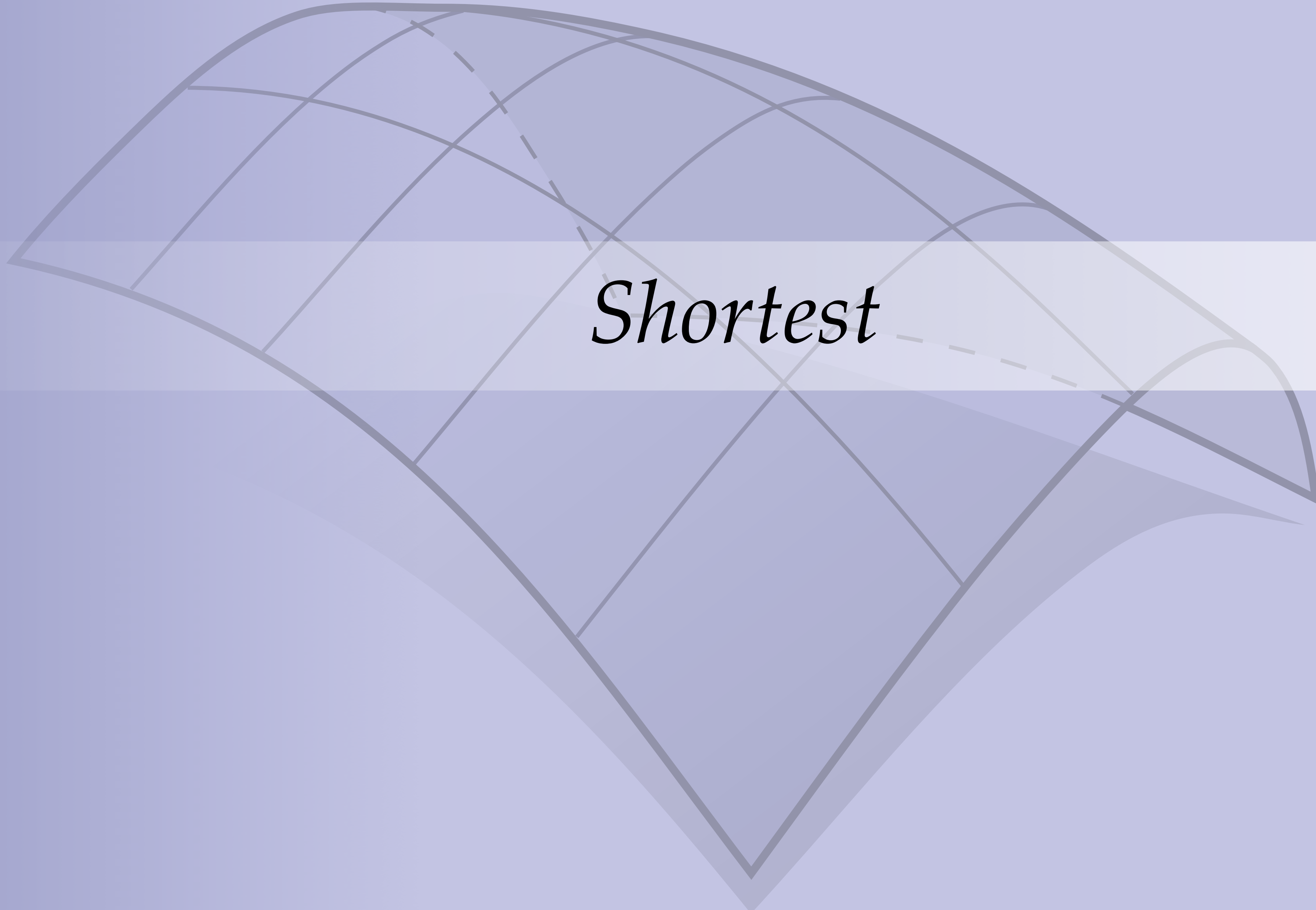
Key fact: geodesics are *isometry invariant*.

Discrete Geodesics

- How can we come up with a definition of *discrete* geodesics?
- Play “The Game” of DDG and consider different smooth starting points:
 - *straightest* (zero acceleration)
 - *locally shortest*
 - *no geodesic curvature*
 - *harmonic map from interval to manifold*
 - *gradient of distance function*
 - ...
- Each starting point will have different consequences



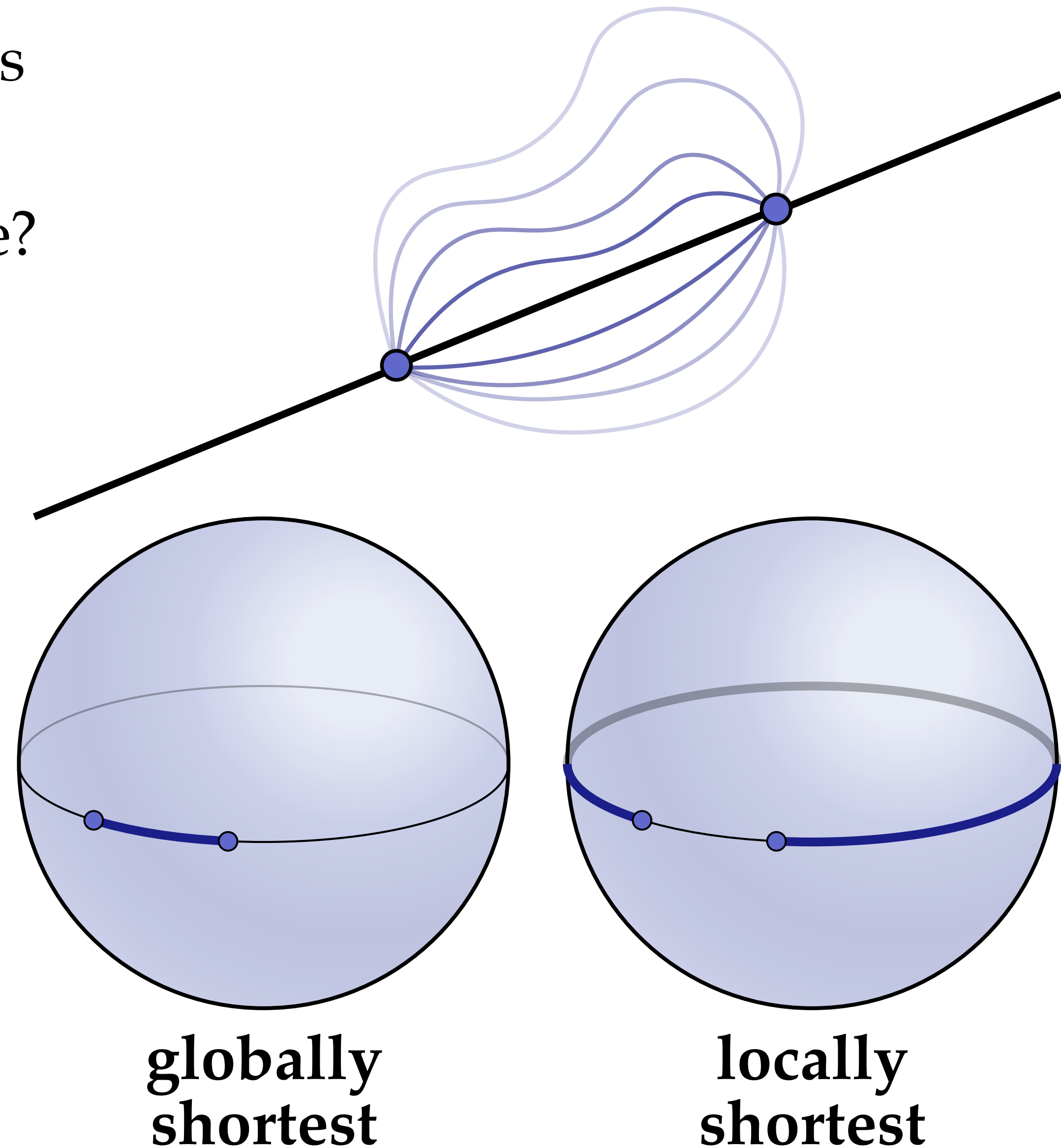
Observation: for simplicial surfaces will see that shortest and straightest disagree



Shortest

Locally Shortest Paths

- A Euclidean line segment can be characterized as the shortest path between two distinct points
- How can we characterize a whole Euclidean line?
 - ...where are the endpoints?
- Say that it's *locally shortest*: for any two “nearby” points on the path, can't find a shorter route
 - “nearby” means shortest path is **unique**
- This description directly gives us one possible definition for geodesics
- Note that *locally* shortest does not imply *globally* shortest!
 - Both are geodesic paths



Dirichlet Energy and Curve Length

Recall *Dirichlet energy*, which measures “smoothness”:

$$\begin{array}{ll} \text{planar curve} & \text{Dirichlet energy} \\ \gamma : [0, 1] \rightarrow \mathbb{R}^2 & E_D(\gamma) = \int_0^1 |\gamma'(t)|^2 dt \end{array}$$

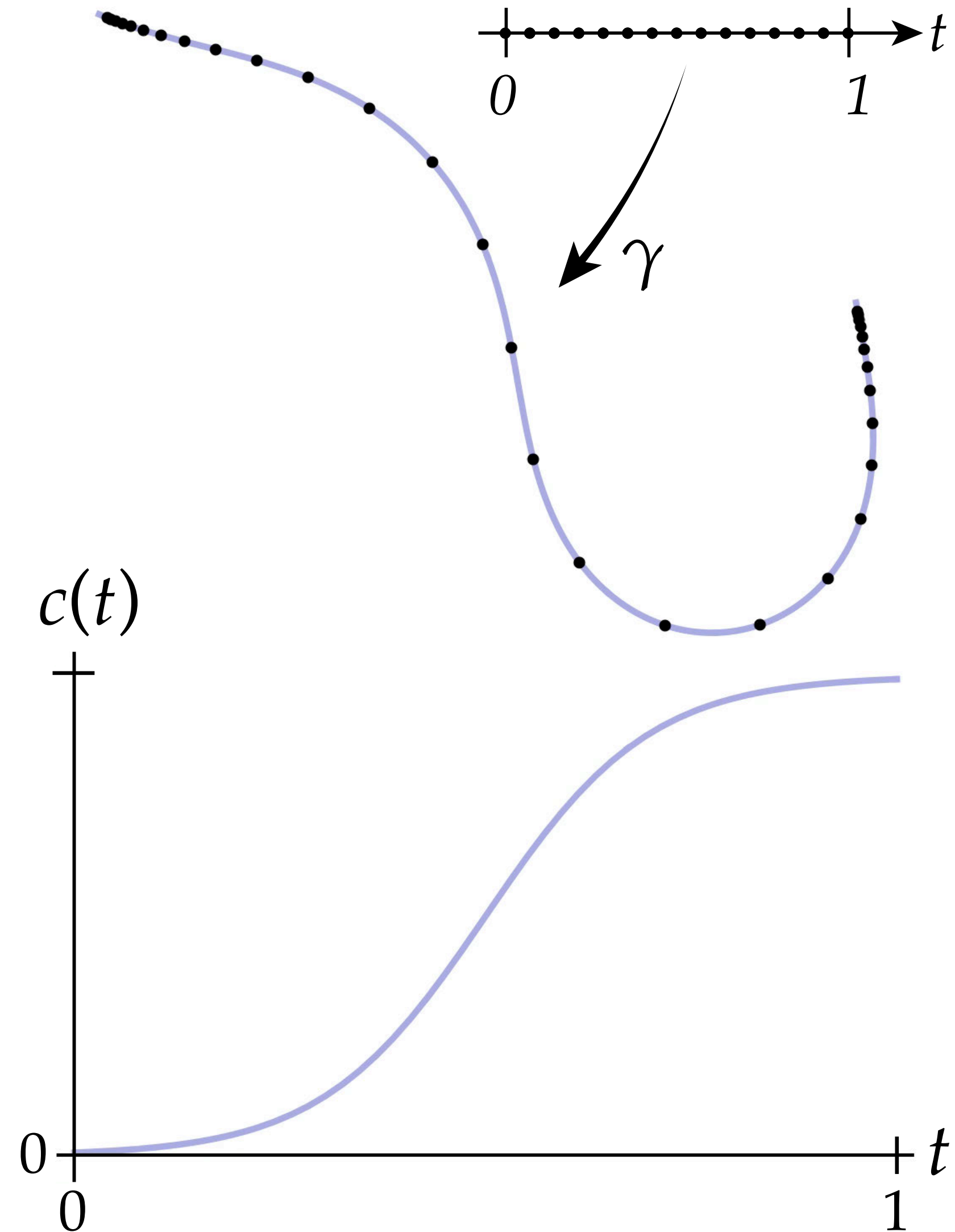
Can write γ as a reparameterization of a unit-speed curve:

$$\begin{array}{ll} \text{unit-speed curve} & \text{speed function} \\ \hat{\gamma} : [0, L] \rightarrow \mathbb{R}^2 & c : [0, 1] \rightarrow \mathbb{R} \\ & c(0) = 0, c(1) = L \end{array} \quad \boxed{\begin{array}{l} \gamma(t) = \hat{\gamma}(c(t)) \\ |\gamma'(t)| = |c'(t)| \end{array}}$$

Now let's try to find the *smoothest* curve...

$$\min_{\gamma} E_D(\gamma) = \min_{\hat{\gamma}} \left(\min_c \int_0^1 \underbrace{(c'(t))^2}_{\rightarrow L} dt \right) = \min_{\hat{\gamma}} L^2$$

Key idea: for a curve, minimizing Dirichlet energy will minimize length.



Shortest Planar Curve—Variational Perspective

- Consider again a curve $\gamma(t): [0,1] \rightarrow \mathbb{R}^2$
- Can find shortest path by minimizing Dirichlet energy, subject to fixed endpoints $\gamma(0)=p$, $\gamma(1)=q$:

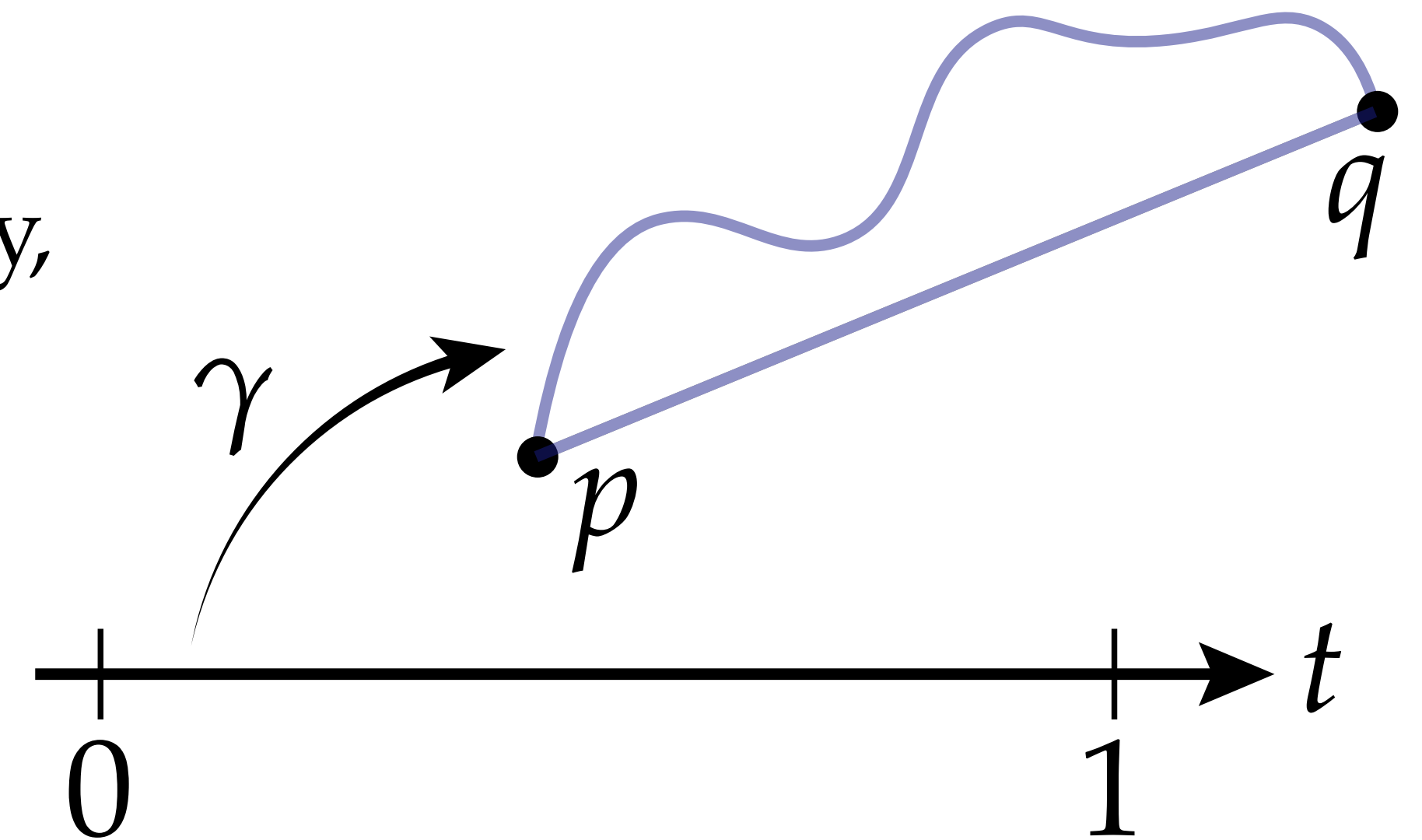
$$\min_{\gamma} \int_0^1 |\gamma'(t)|^2 dt$$

(integration by parts)

$$\iff \min_{\gamma} - \int_0^1 \langle \gamma(t), \gamma''(t) \rangle dt$$

- Taking gradient w.r.t. γ yields a 1D Poisson equation
- **Q:** Solution? **A:** Linear function!

Key idea: geodesics are harmonic functions



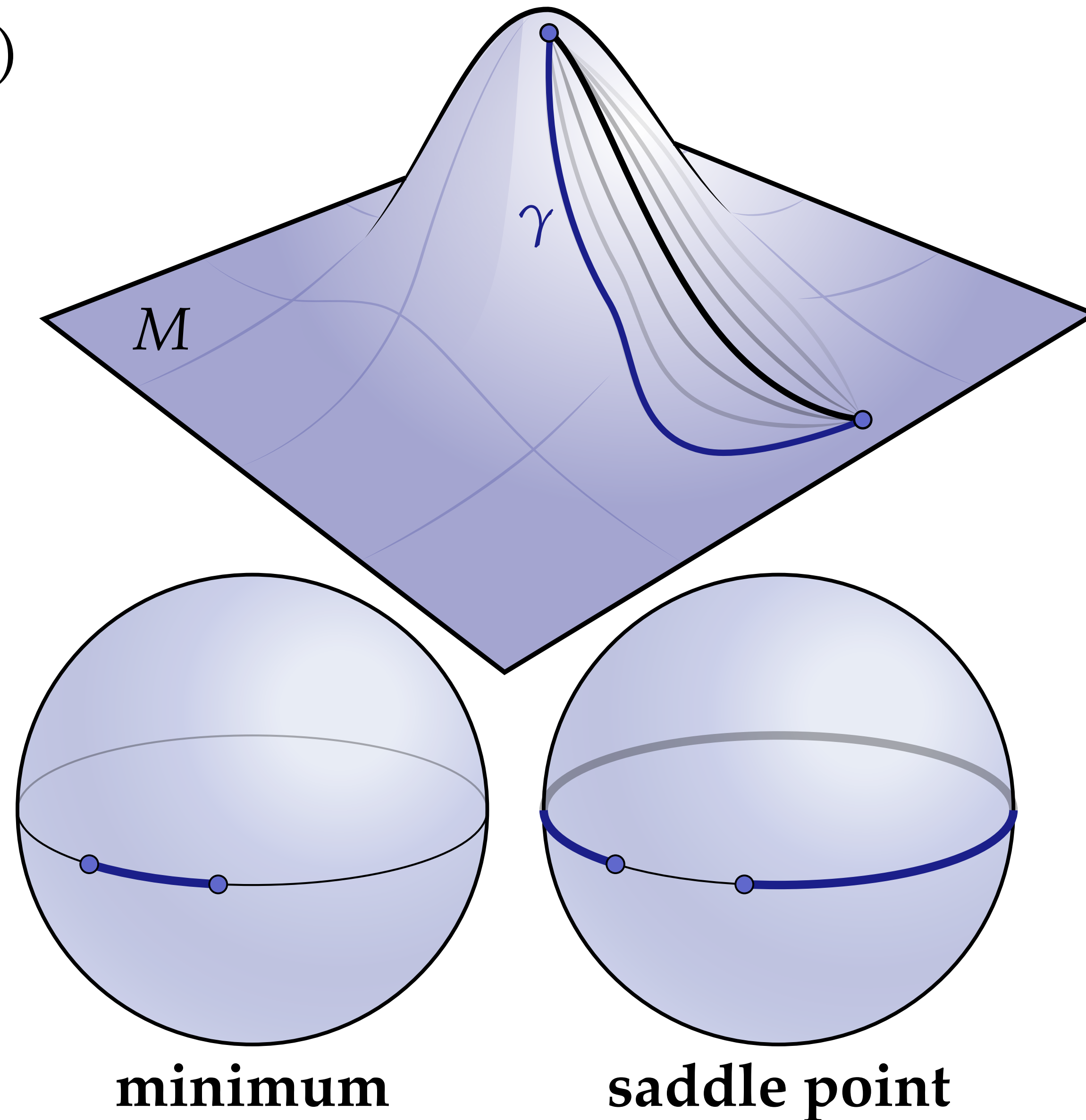
$\frac{\partial^2}{\partial t^2} \gamma(t)$	$=$	0
$\gamma(0)$	$=$	p
$\gamma(1)$	$=$	q

Shortest Geodesic—Variational Perspective

- Essentially same story on a curved surface (M,g)
- Consider a differentiable curve $\gamma: [0,1] \rightarrow M$
- Dirichlet energy is then

$$E_D(\gamma) = \int_0^1 |\gamma'(t)|^2 dt = \int_0^1 g(\gamma'(t), \gamma'(t)) dt$$

- Geodesics are still critical points (*harmonic*)
- May no longer be global minimizers

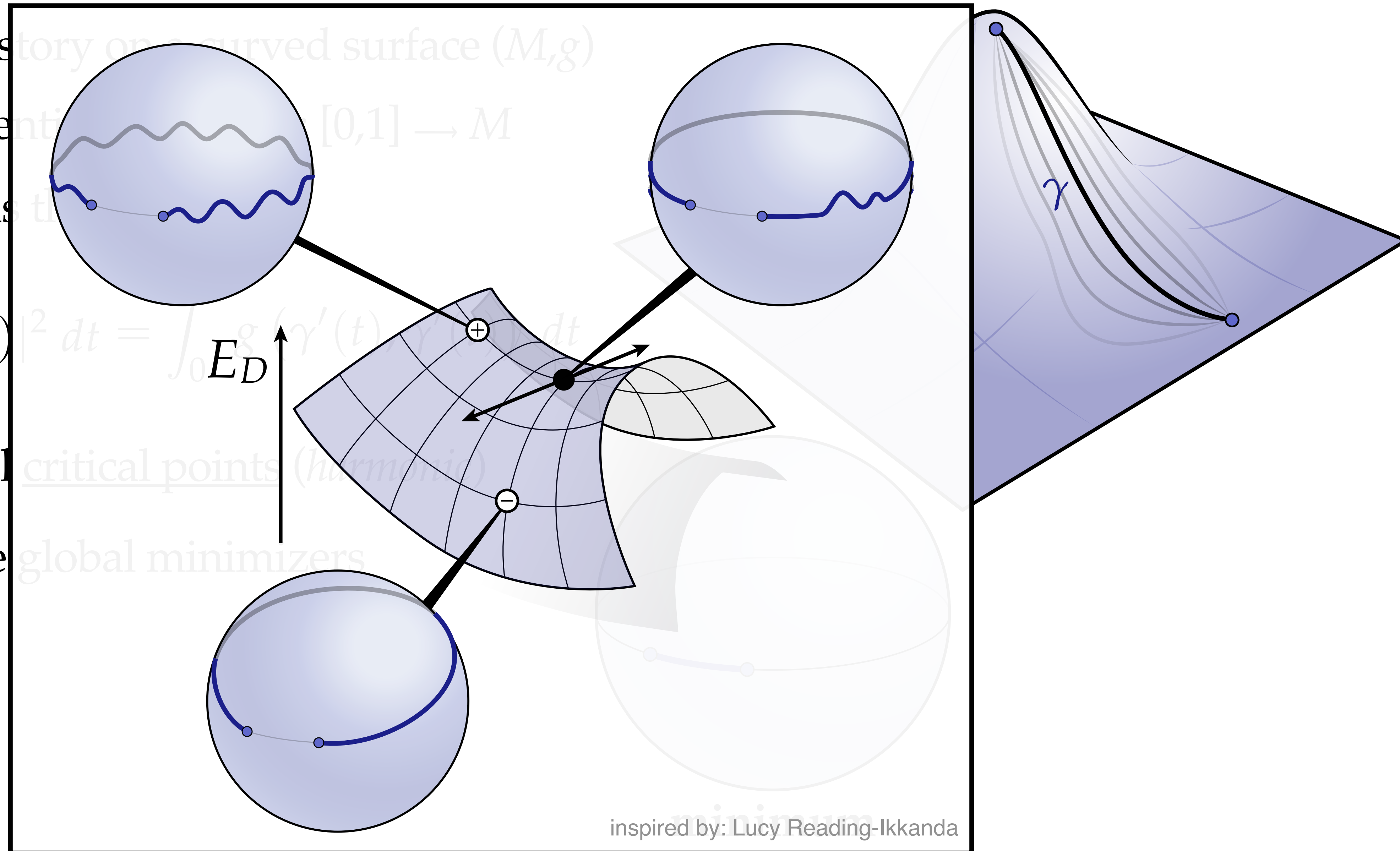


Shortest Geodesic—Variational Perspective

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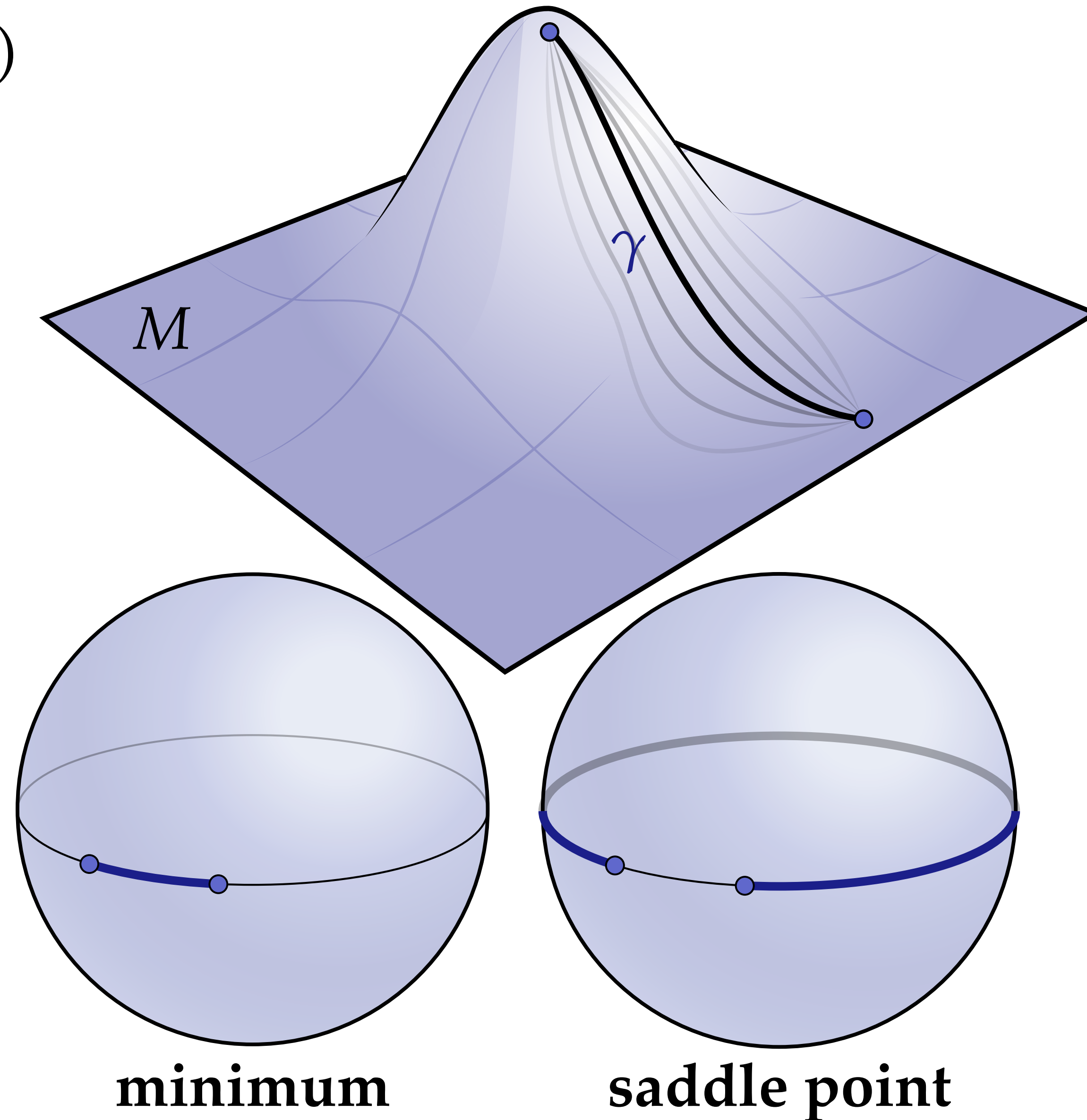


Shortest Geodesic—Variational Perspective

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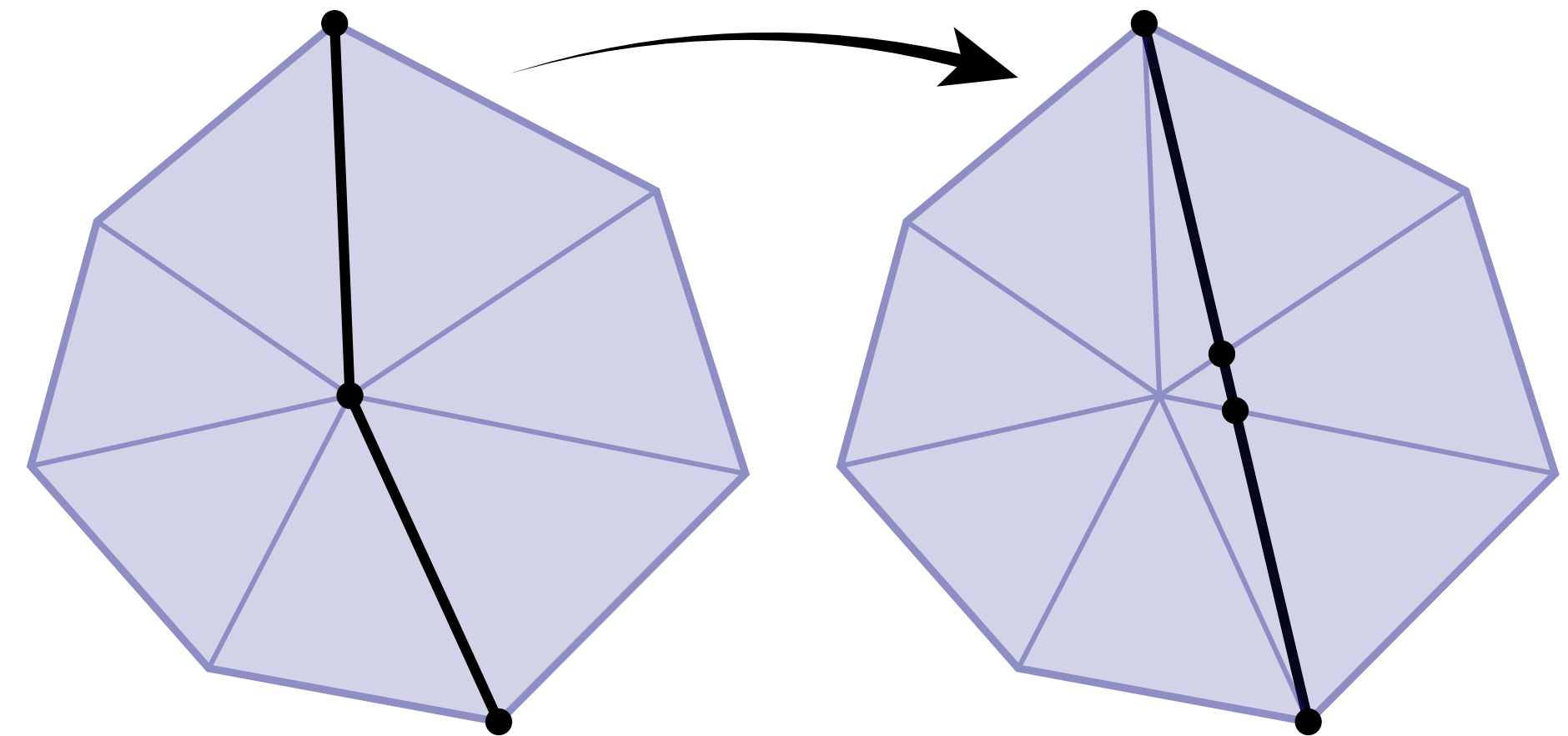
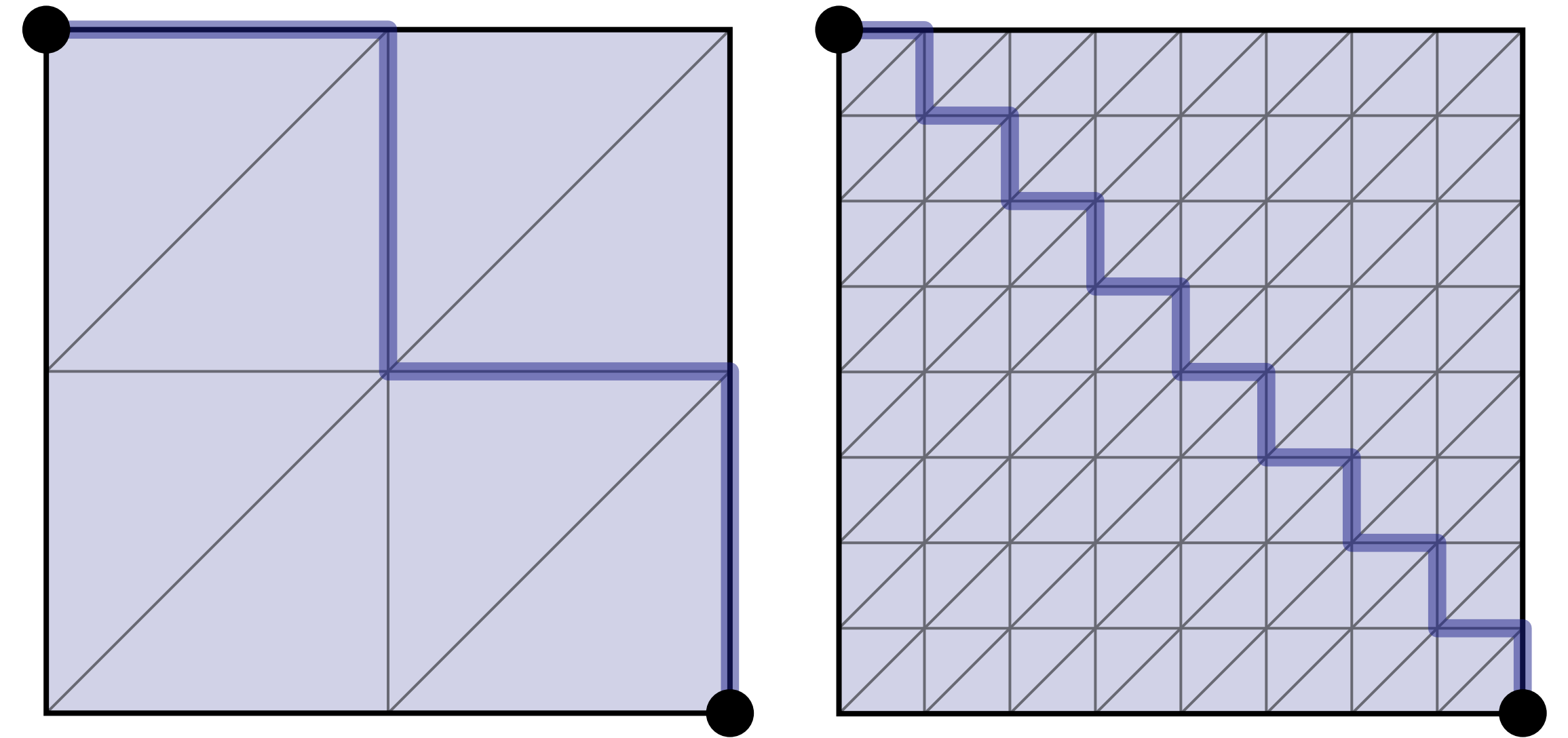
$$E_D(\gamma) = \int_0^1 |\gamma'(t)|^2 dt = \int_0^1 g(\gamma'(t), \gamma'(t)) dt$$

- Geodesics are still critical points (*harmonic*)
- May no longer be global minimizers
- Hence, geodesics no longer found by solving easy linear equation (Laplace)
 - Will need *numerical algorithms*!



Discrete Shortest Paths—Boundary Value Problem

- **Q:** How can we find a shortest path in the discrete case?
- Dijkstra's algorithm obviously comes to mind, but a shortest path in the edge graph is almost never geodesic
 - even if you refine the mesh!
- To get *locally* shortest path, could iteratively straighten Dijkstra path by until no more progress can be made
- What if we want to compute the *globally* shortest path?

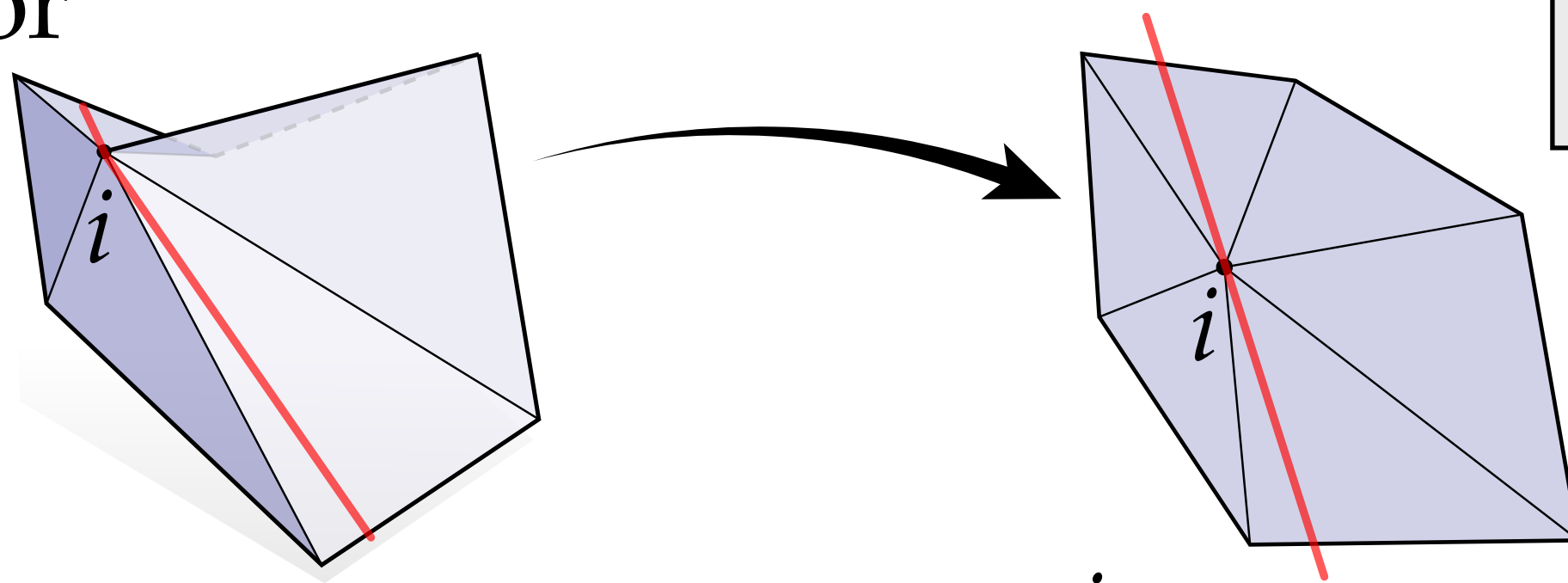


Discrete Shortest Paths—Vertices

- Even *locally* shortest paths near vertices require some care—different behavior depending on angle defect Ω

- Flat ($\Omega = 0$)

Shortest path simply goes straight through the vertex

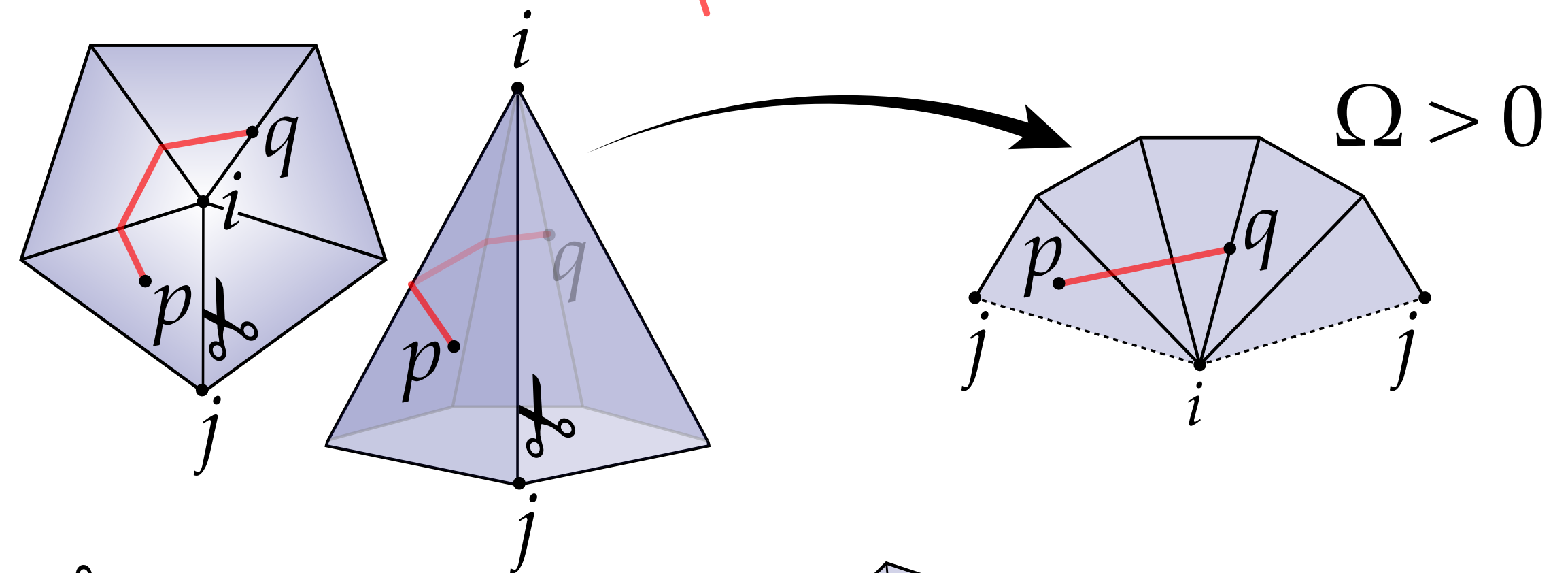


$$\Omega_i = 2\pi - \sum_{ijk} \theta_i^{jk}$$

$\Omega = 0$

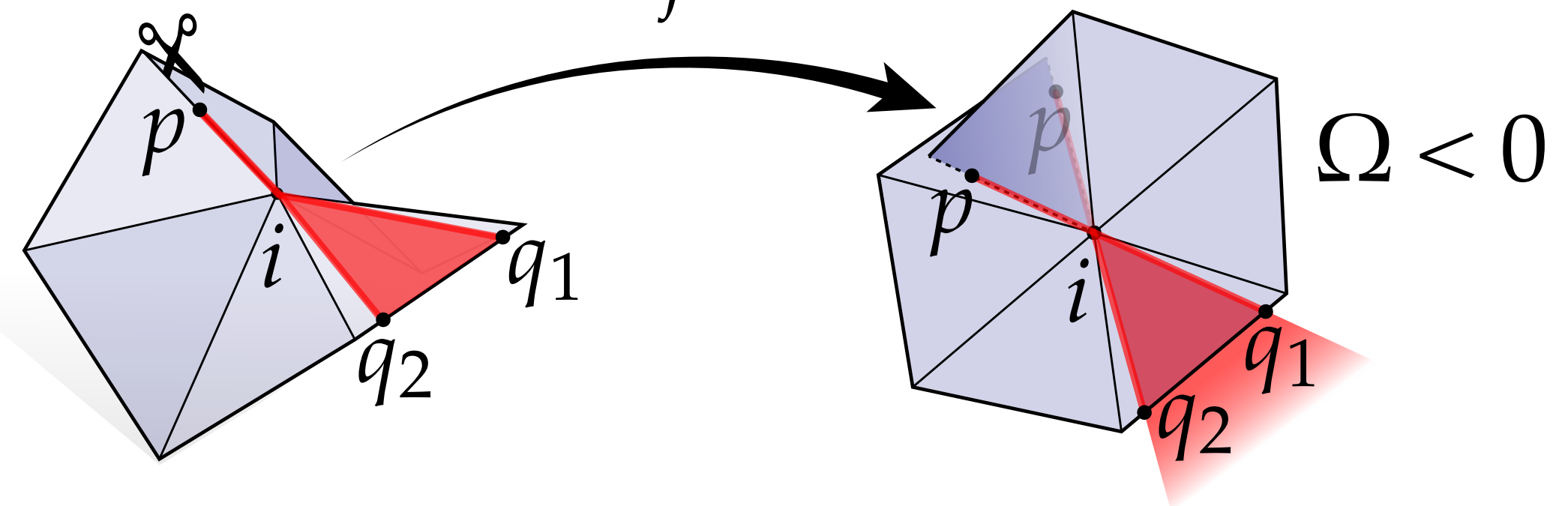
- Cone ($\Omega > 0$)

Can always faster to go around one side or the other; never *through* the vertex



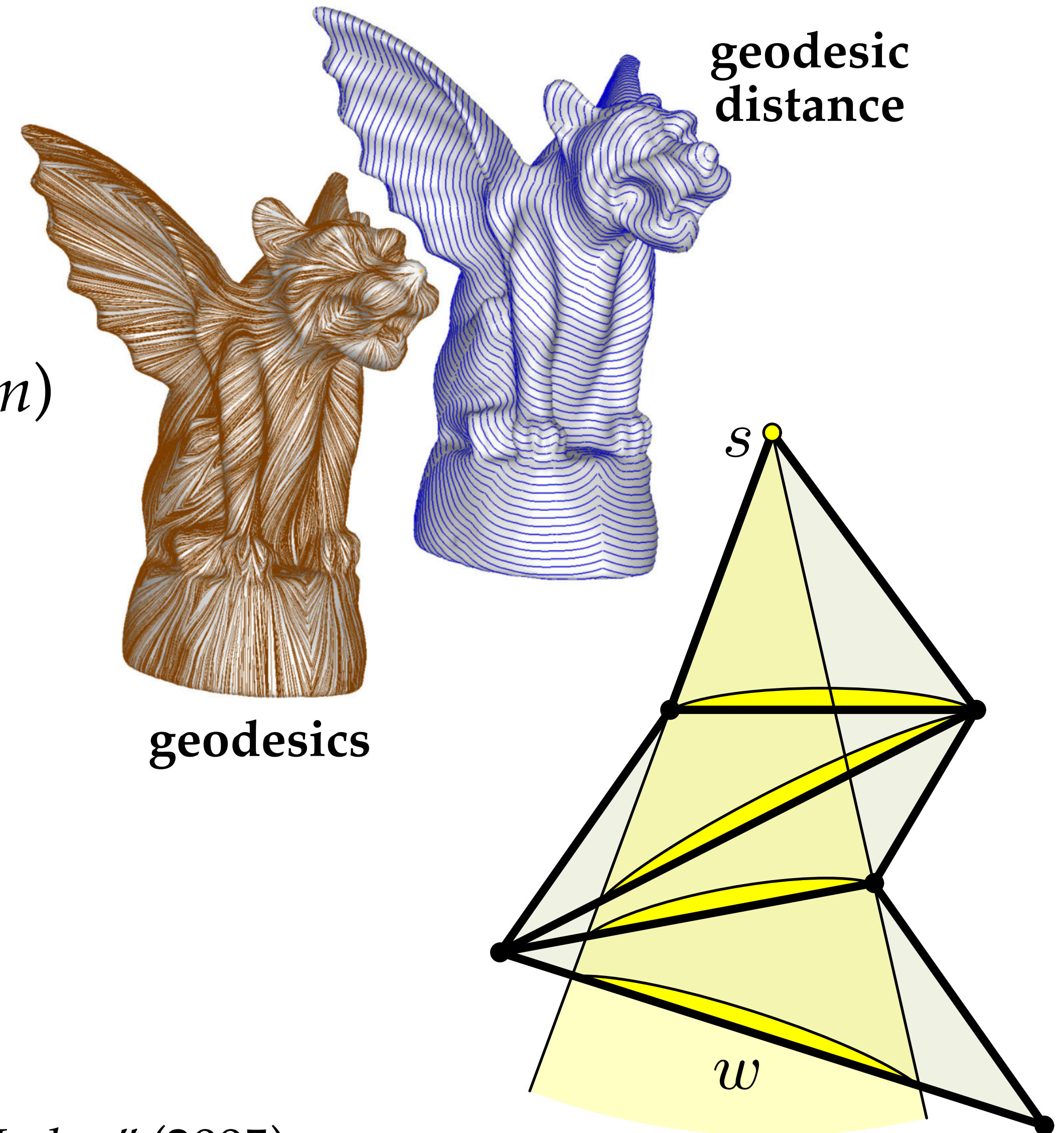
- Saddle ($\Omega < 0$)

Always *many* locally shortest paths passing through a saddle vertex.



Algorithms for Shortest Polyhedral Geodesics

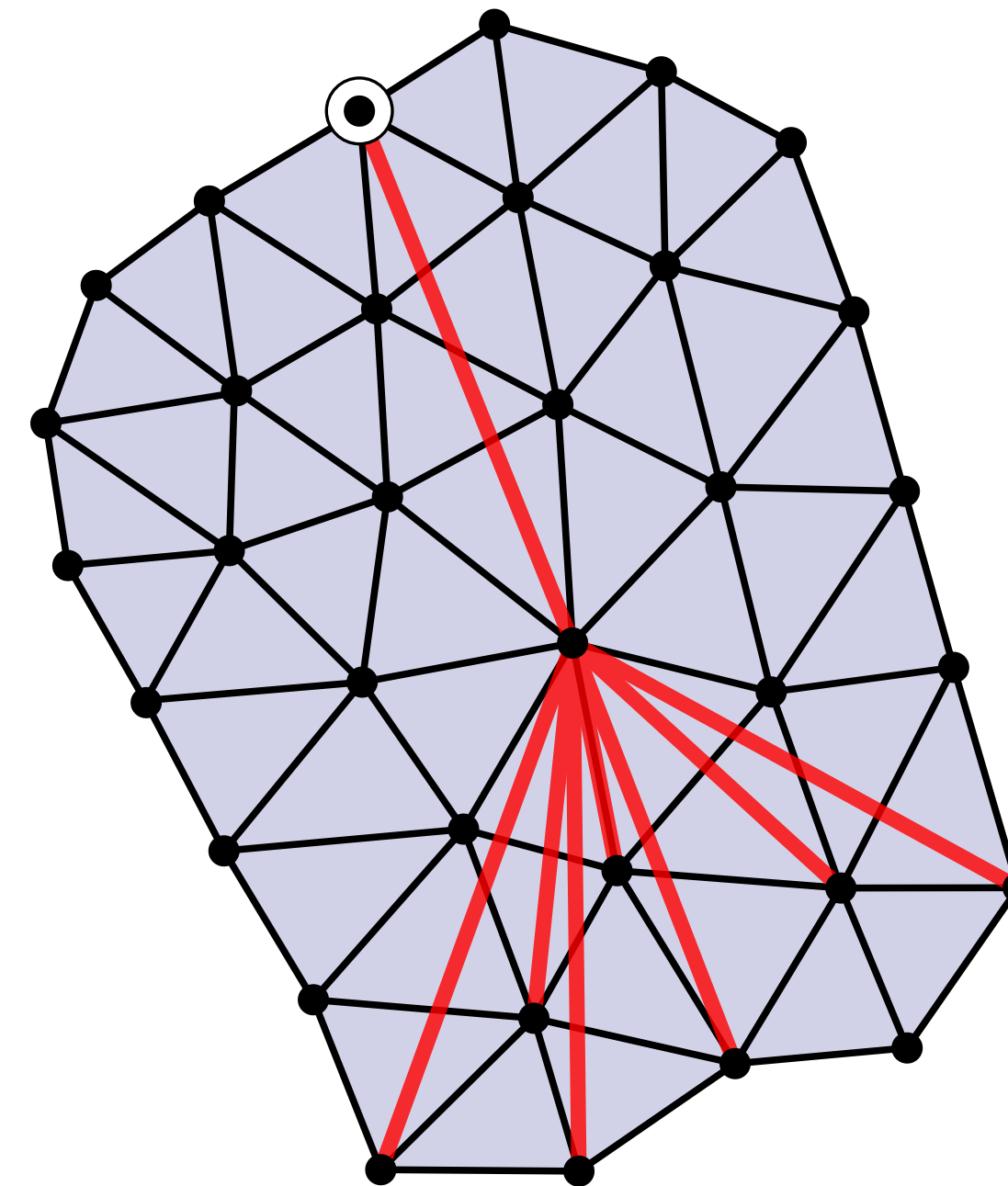
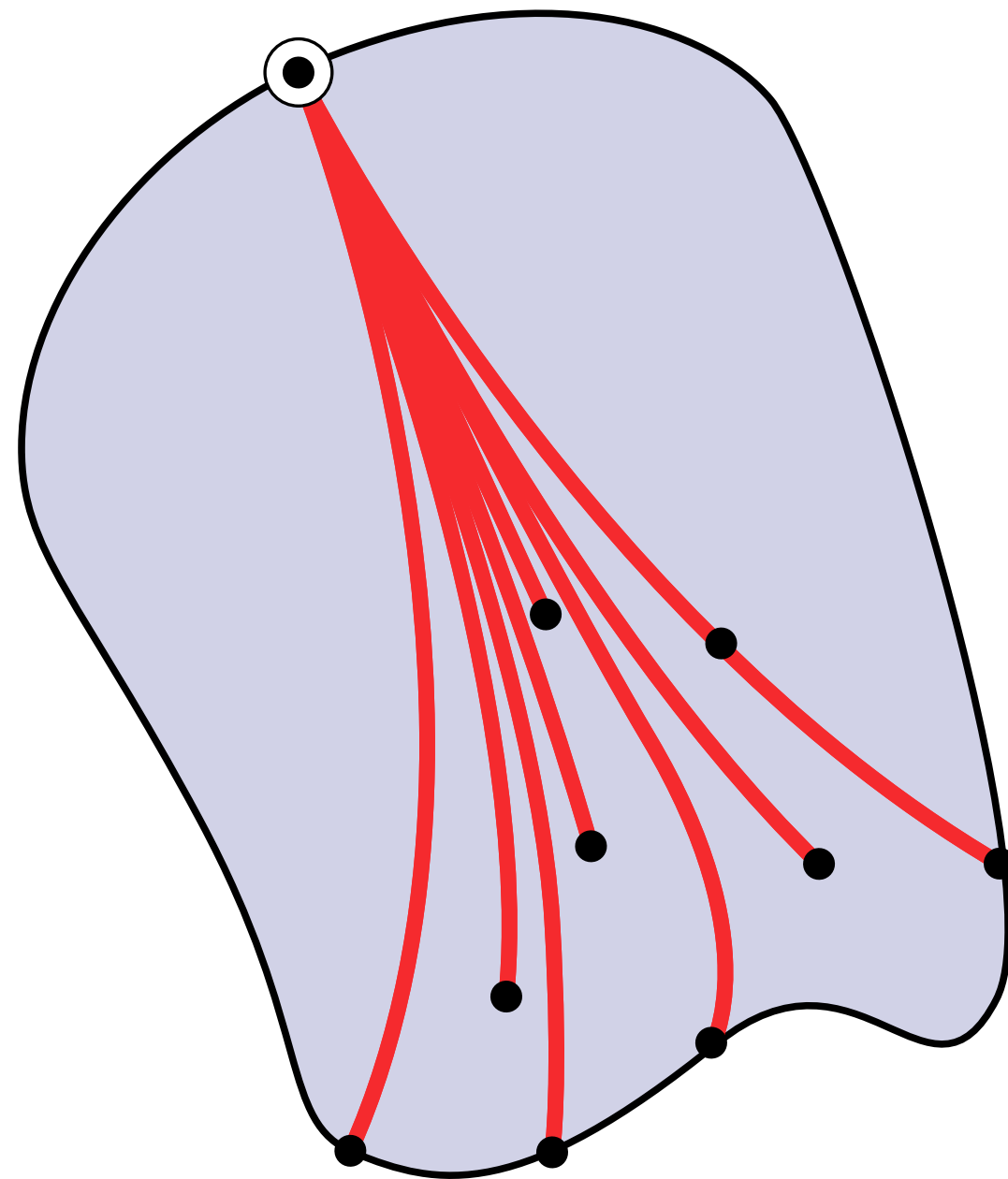
- Algorithms for *shortest* polyhedral geodesics generalize Dijkstra's algorithm to include paths through triangles
- Mitchell, Mount, Papadimitrou (MMP) "*The Discrete Geodesic Problem*" (1986) — $O(n^2 \log n)$
- **Basic idea:** track intervals or "windows" of common geodesic paths
- Many subsequent improvements by pruning windows, approximation, ... though still quite expensive (same asymptotic complexity)



See: Surazhsky et al. "*Fast Exact and Approximate Geodesics on Meshes*" (2005)

Shortest Geodesics — Smooth vs. Discrete

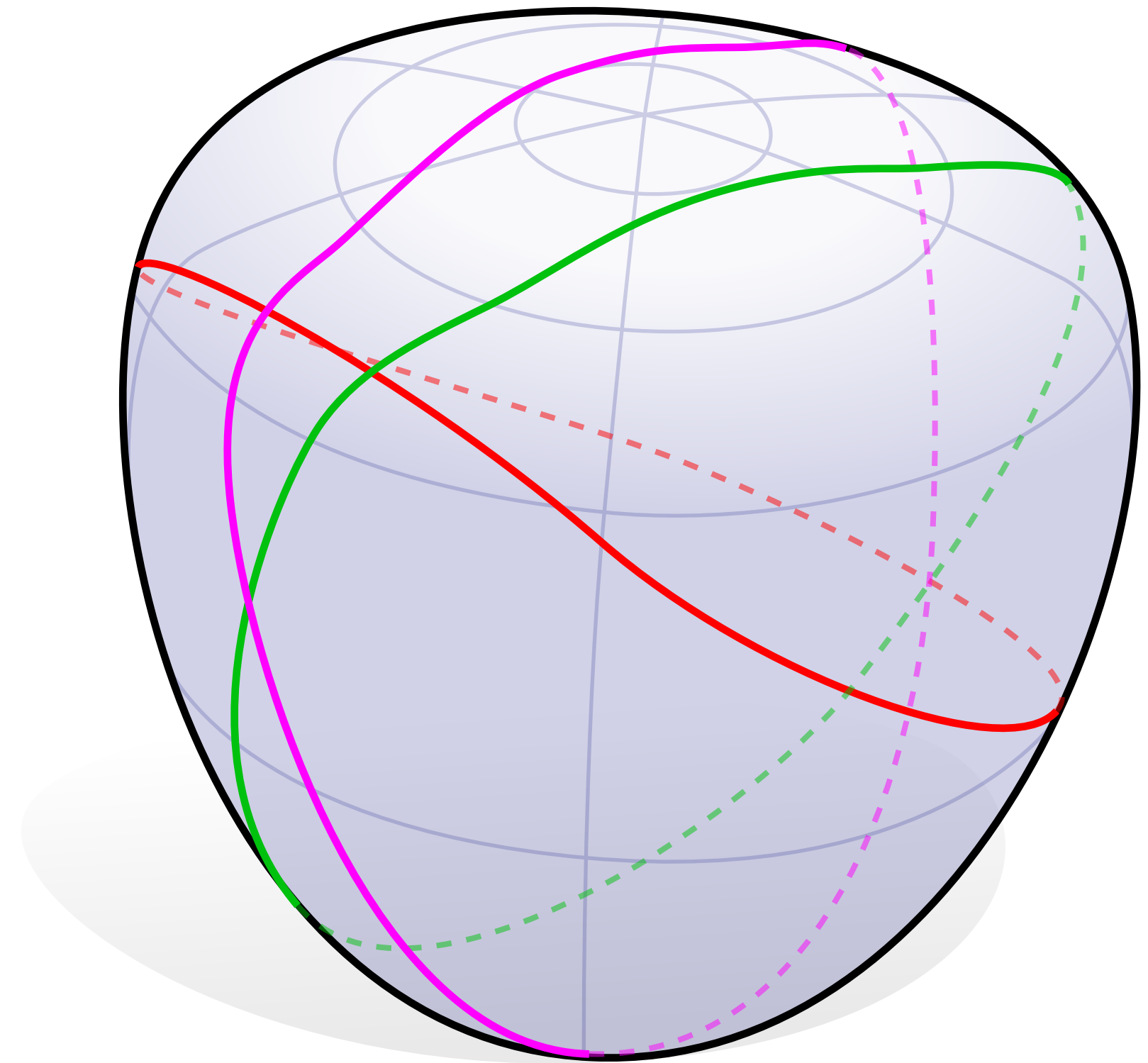
- **Smooth:** two minimal geodesics γ_1, γ_2 from a source p to distinct points p_1, p_2 (resp.) intersect only if $\gamma_1 \subseteq \gamma_2$ or $\gamma_2 \subseteq \gamma_1$
- **Discrete:** many geodesics can coincide at saddle vertex (“pseudo-source”)



Note: Shortest polyhedral geodesics may not faithfully capture behavior of smooth ones!

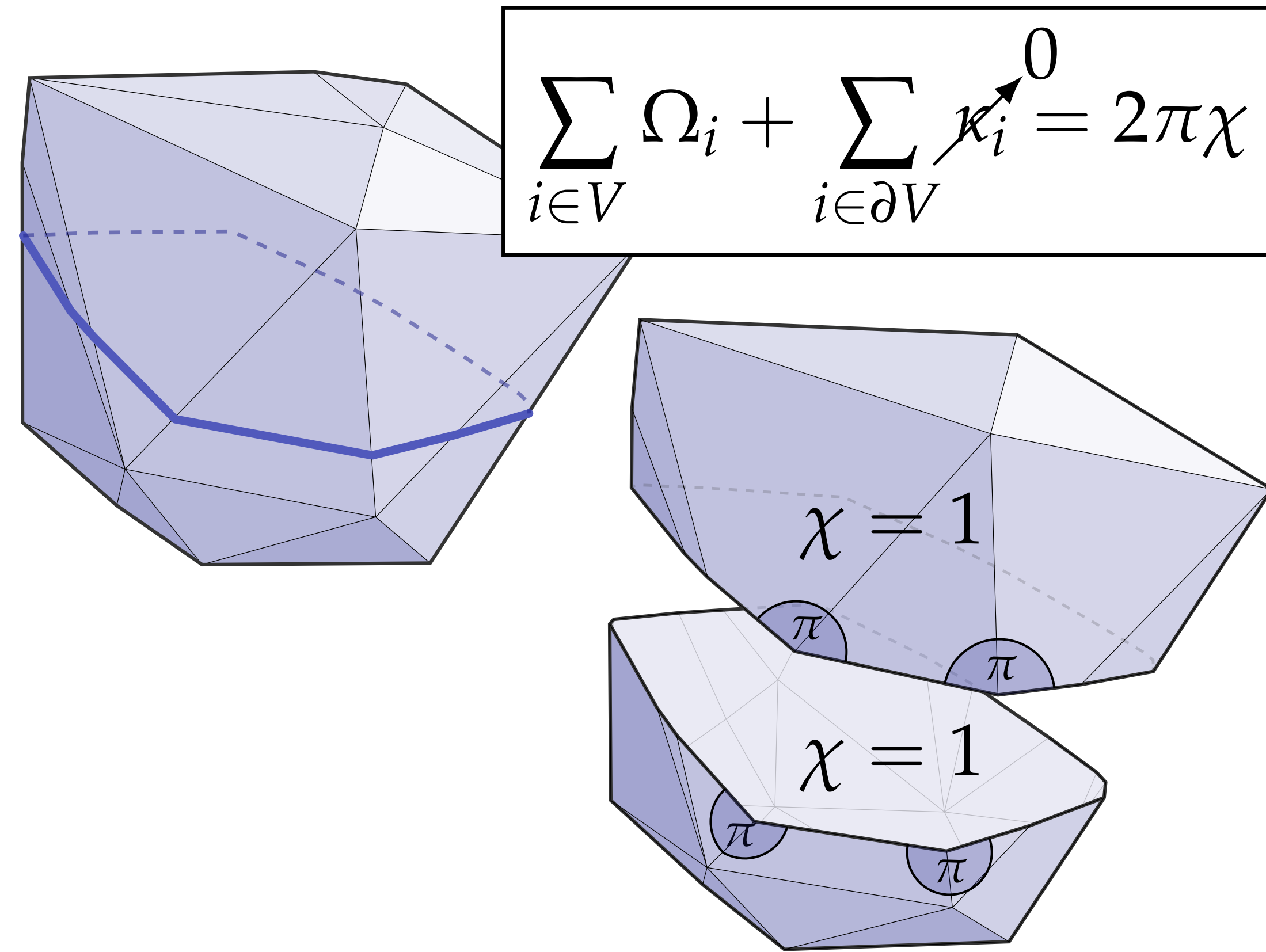
Closed Geodesics

- **Theorem.** (Birkhoff 1917) Every smooth convex surface contains a simple closed geodesic, *i.e.*, a geodesic loop that does not cross itself (“*Birkhoff equator*”)
- **Theorem.** (Luysternik & Shnirel’man 1929) Actually, there are at least three—and this result is sharp: *only* three on some smooth surfaces.



Closed Geodesics

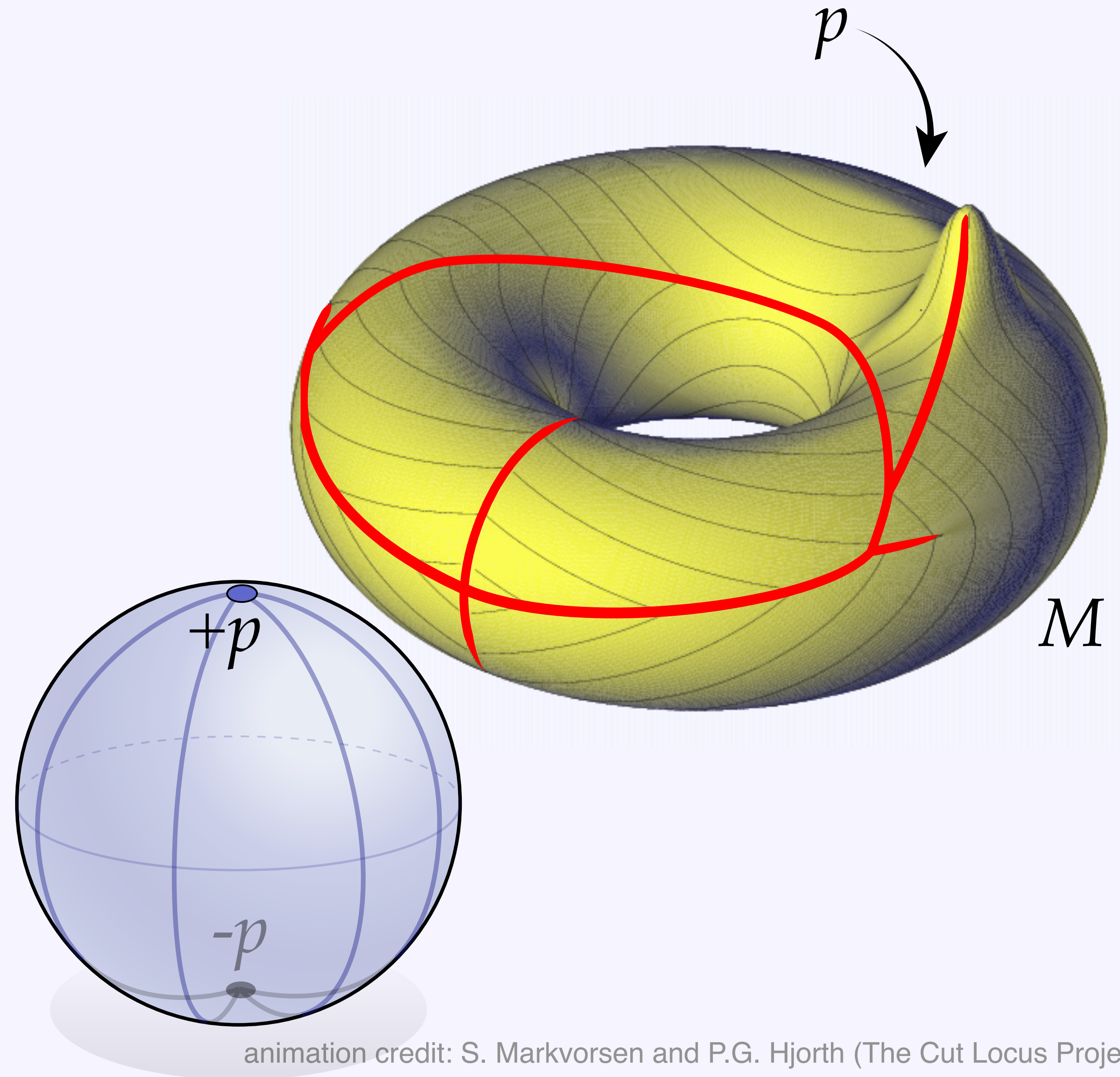
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- **Theorem.** (Luysternik & Shnirel’man 1929) Actually, there are at least three—and this result is sharp: *only* three on some smooth surfaces.
- **Theorem.** (Galperin 2002) *Most* convex polyhedra do not have simple closed geodesics (in the sense of discrete *shortest* geodesics).
- *Shortest* characterization of discrete geodesics again fails to capture properties from smooth setting.



A *shortest* geodesic can't pass through convex vertices. So, by Gauss-Bonnet, a closed geodesic would have to partition vertices into two sets that each have total angle defect of exactly 2π .

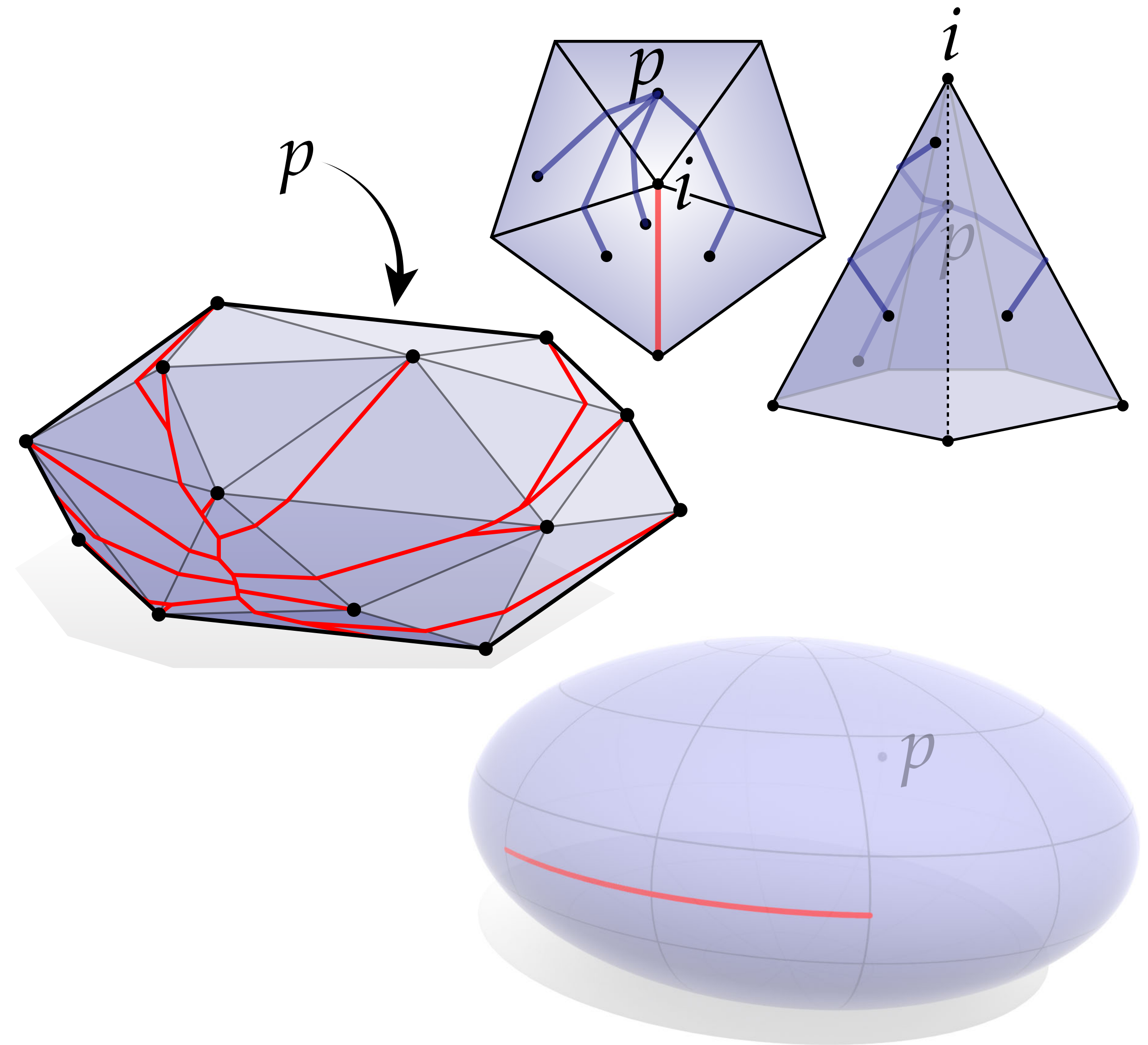
Cut Locus & Injectivity Radius

- For a source point p on a smooth surface M , the *cut locus* is the set of all points q such that there is not a unique (globally) shortest geodesic between p and q .
 - *injectivity radius* is the distance to the closest point on the cut locus
- E.g., on a sphere cut locus of any point $+p$ is the antipodal point $-p$
 - injectivity radius covers whole sphere
- In general can be *much* more complicated (and smaller injectivity radius...)



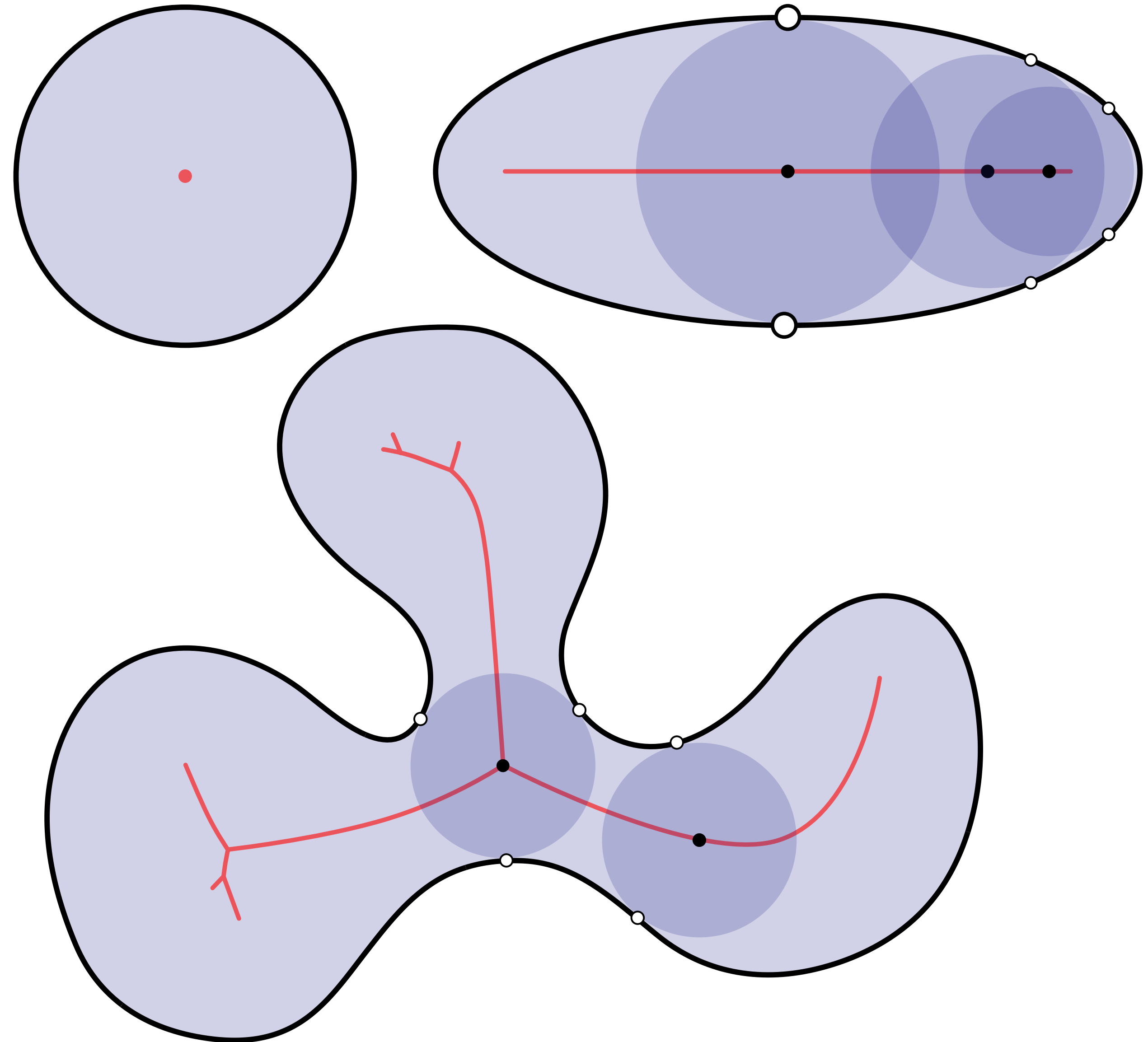
Discrete Cut Locus

- What does cut locus look like for polyhedral surfaces?
- Recall that it's always shorter to go “around” a cone-like vertex (i.e., vertex with positive curvature $\Omega_i > 0$)
- Hence, polyhedral cut locus will contain every cone vertex in the entire surface
- Can look *very* different from the smooth cut locus!



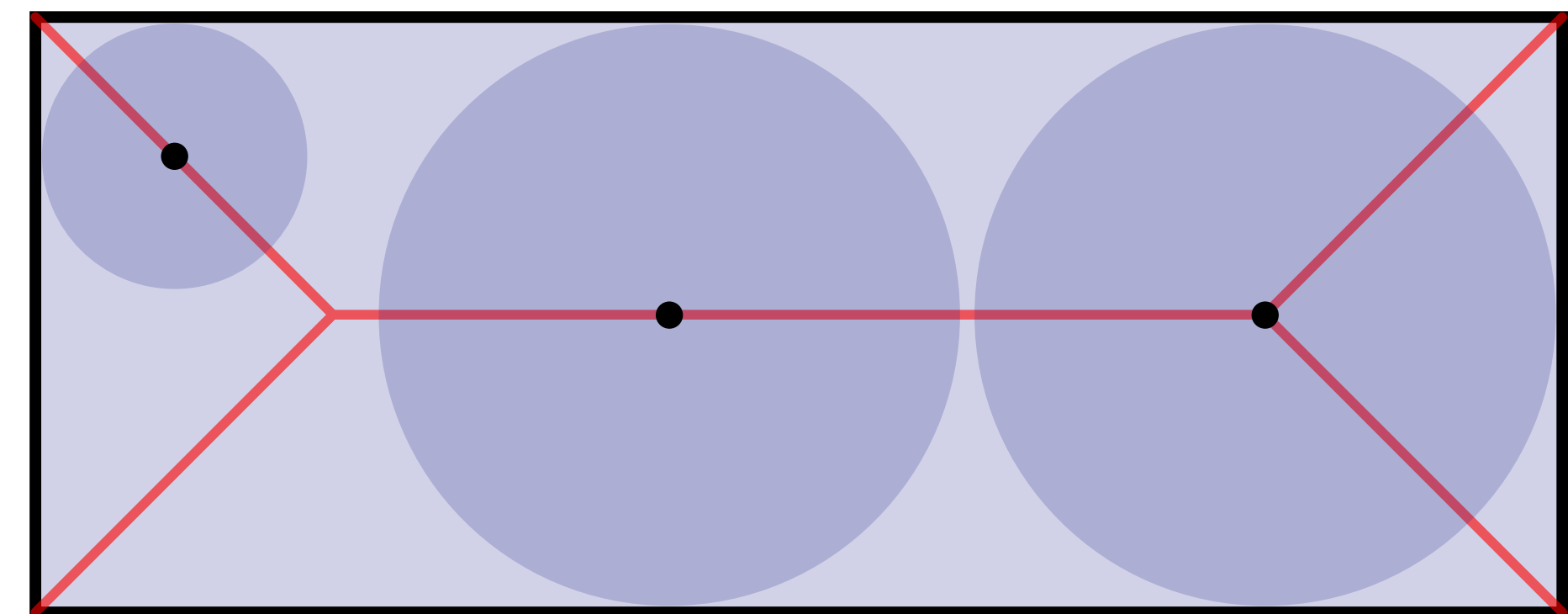
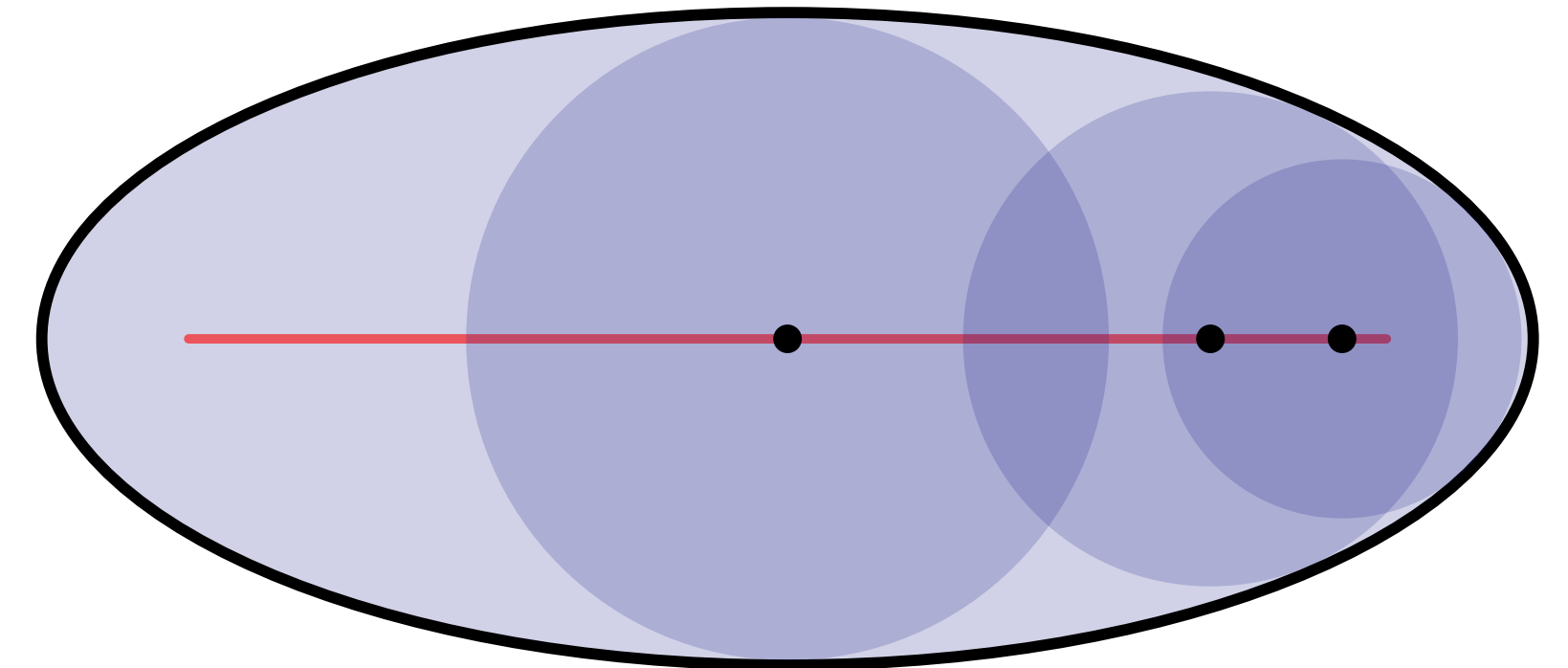
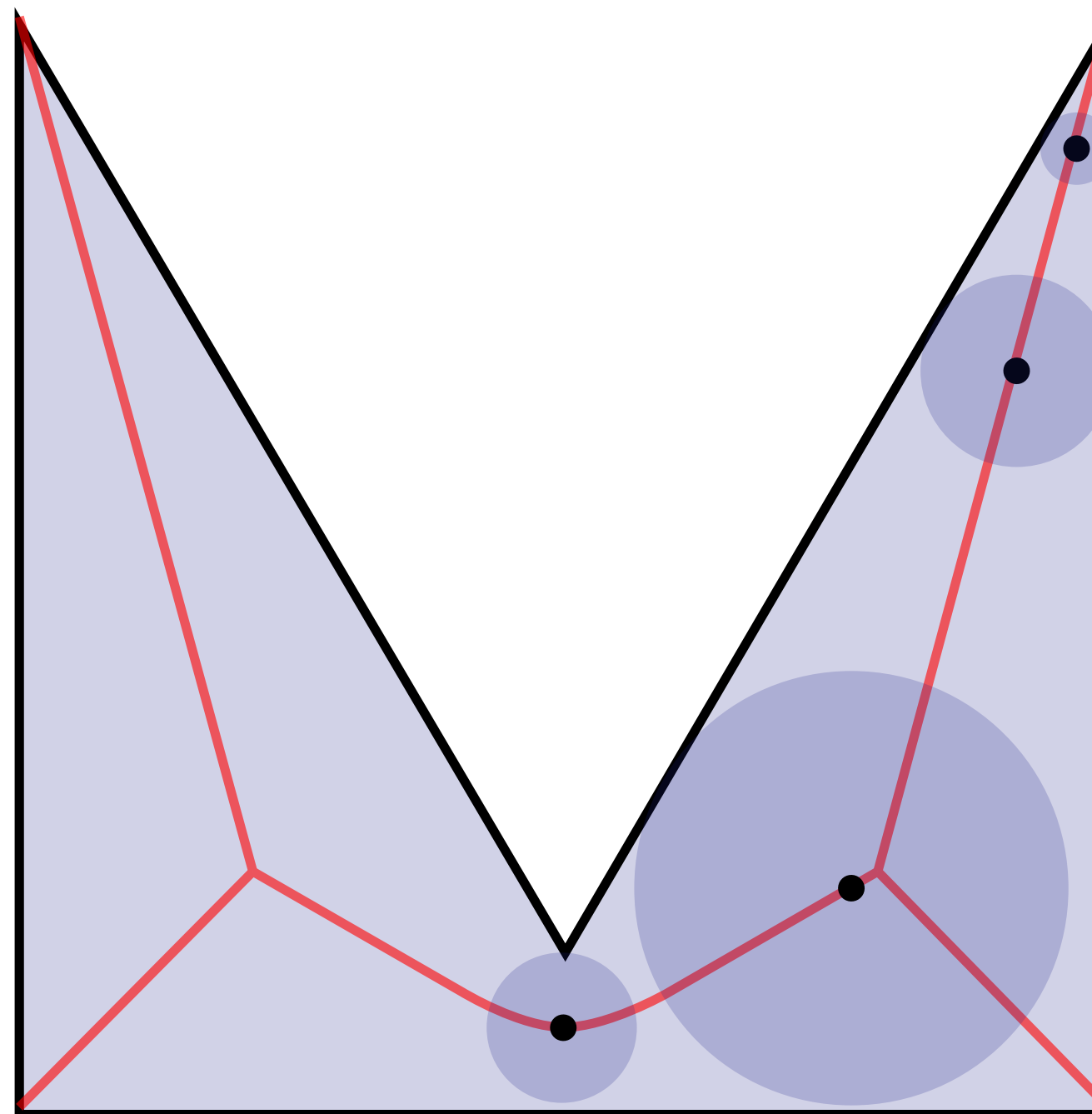
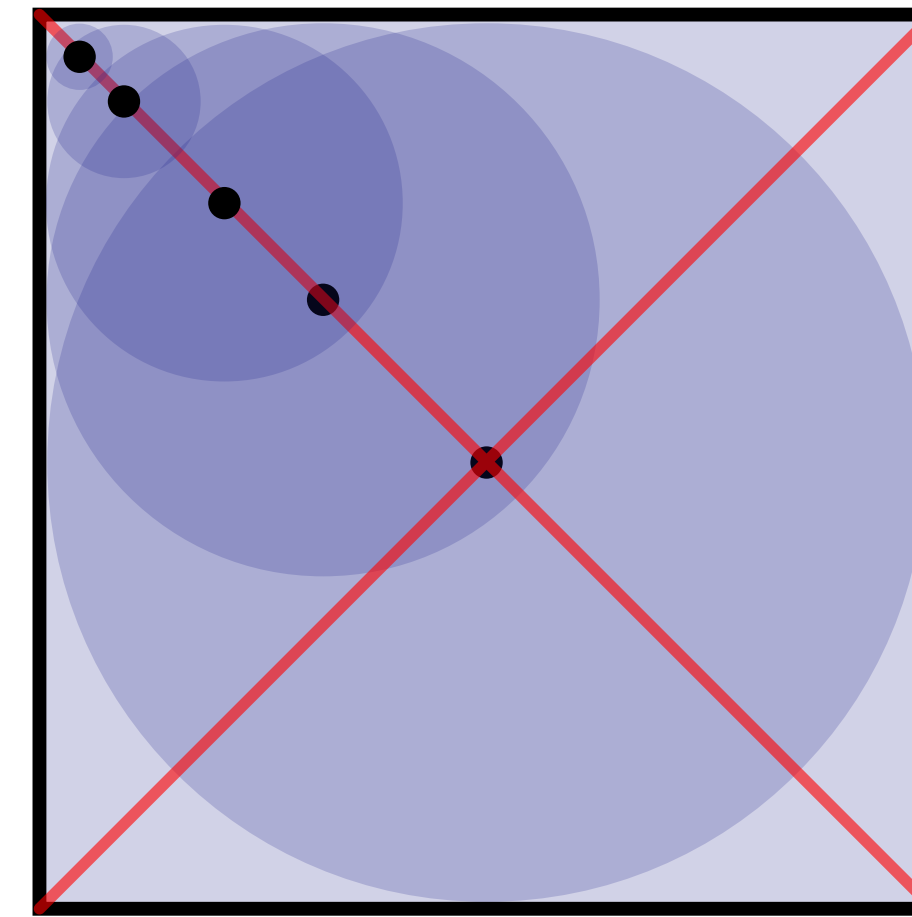
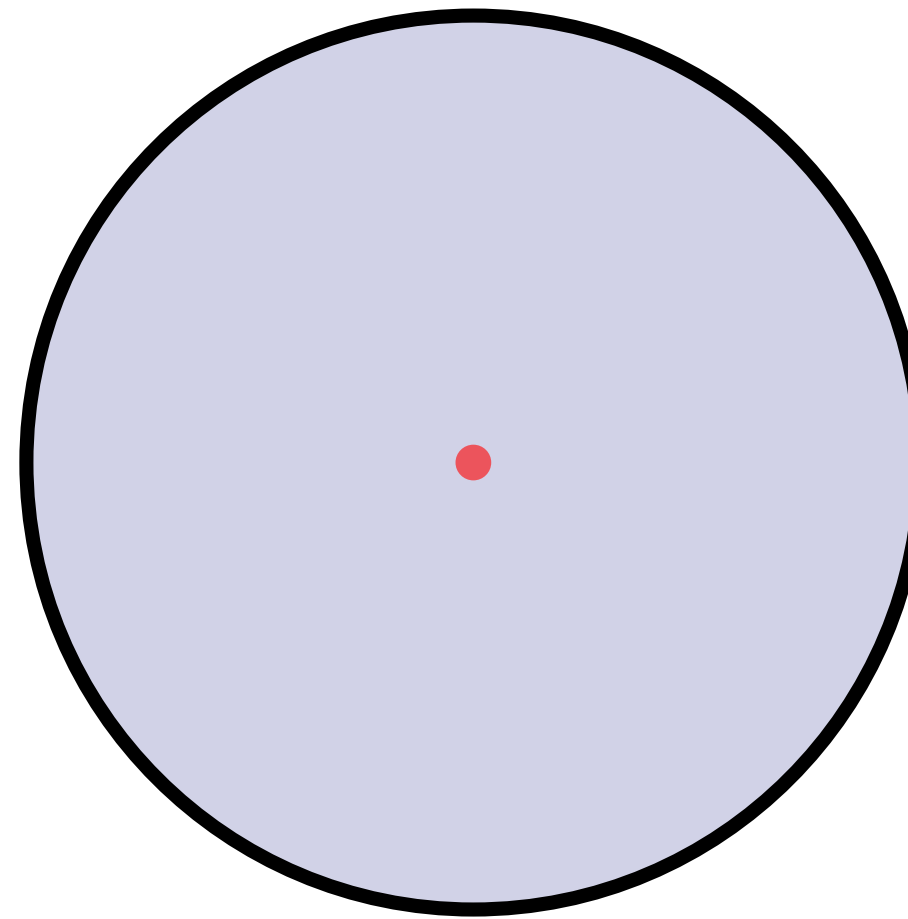
Medial Axis

- Similar to the cut locus, the *medial axis* of a surface or region is the set of all points p that do not have a unique closest point on the boundary
- A *medial ball* is a ball with center on the medial axis, and radius given by the distance to the closest point
- Like cut locus, can get quite complicated!
- Typically three branches (*why?*)
- Provides a “dual” representation: can recover original shape from
 - medial axis
 - radius function



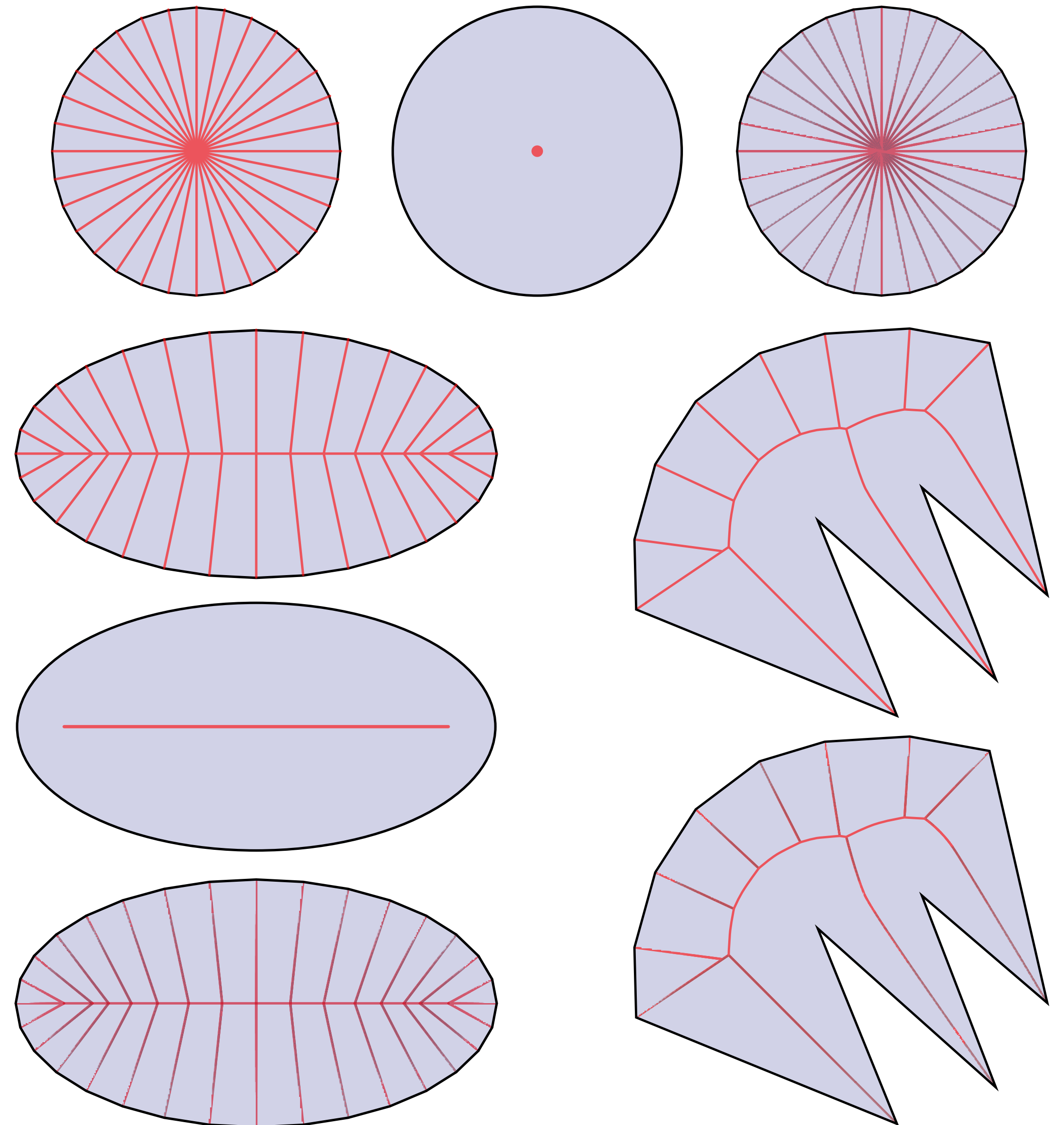
Discrete Medial Axis

- What does the medial axis of a discrete domain look like?
- Let's start with a square.
(What did the medial axis for a circle look like?)
- What about a rectangle?
(What happened with an ellipse?)
- How about a nonconvex polygon?
 - *surprise*: no longer just straight edges!



Discrete Medial Axis

- In general, medial axis touches *every* convex vertex
- May not look much like true (smooth) medial axis!
- One idea: “filter” using radius function...
 - still hard to say exactly which pieces should remain
- Lots of work on alternative “*shape skeletons*” for discrete curves & surfaces



Medial Axis in 3D

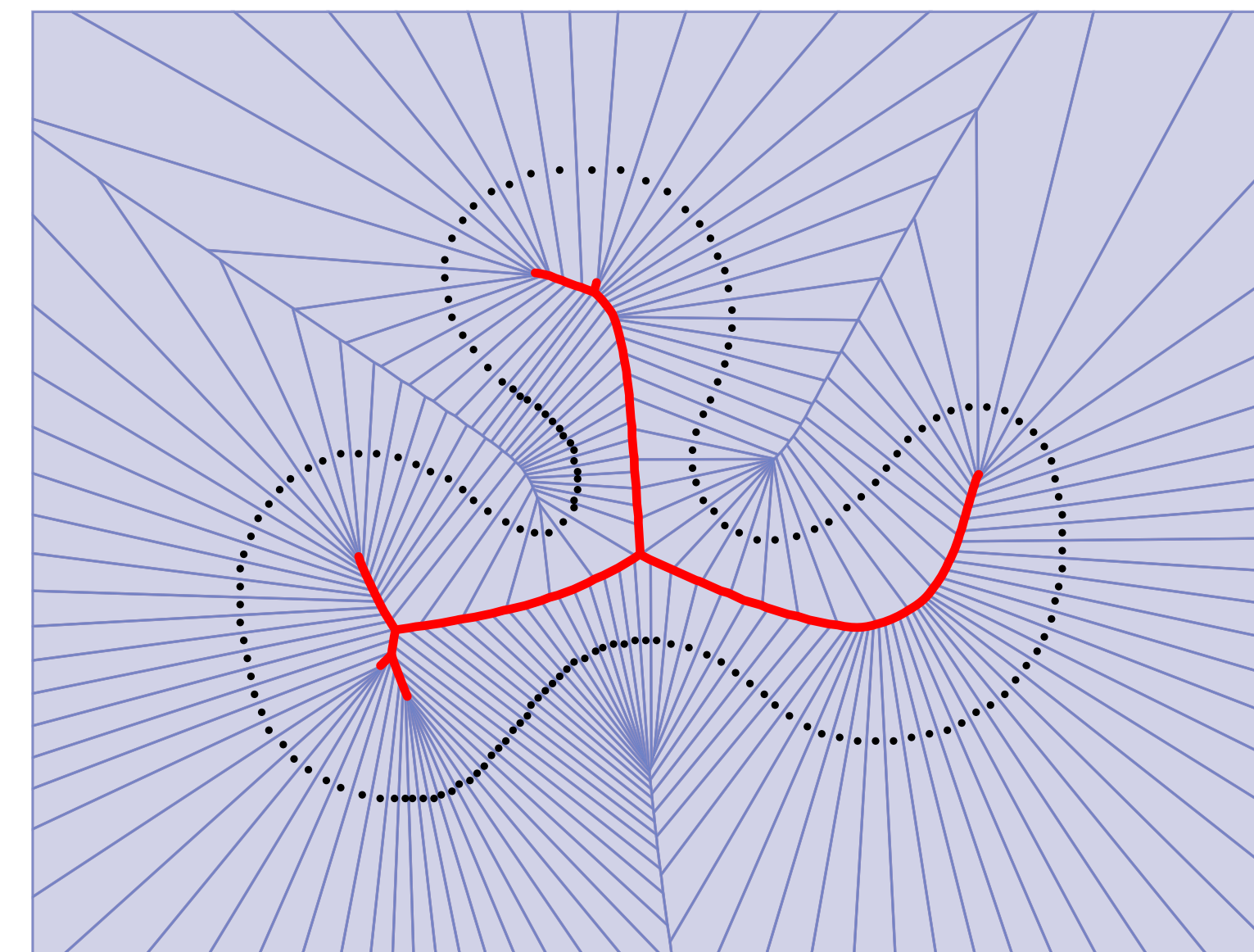
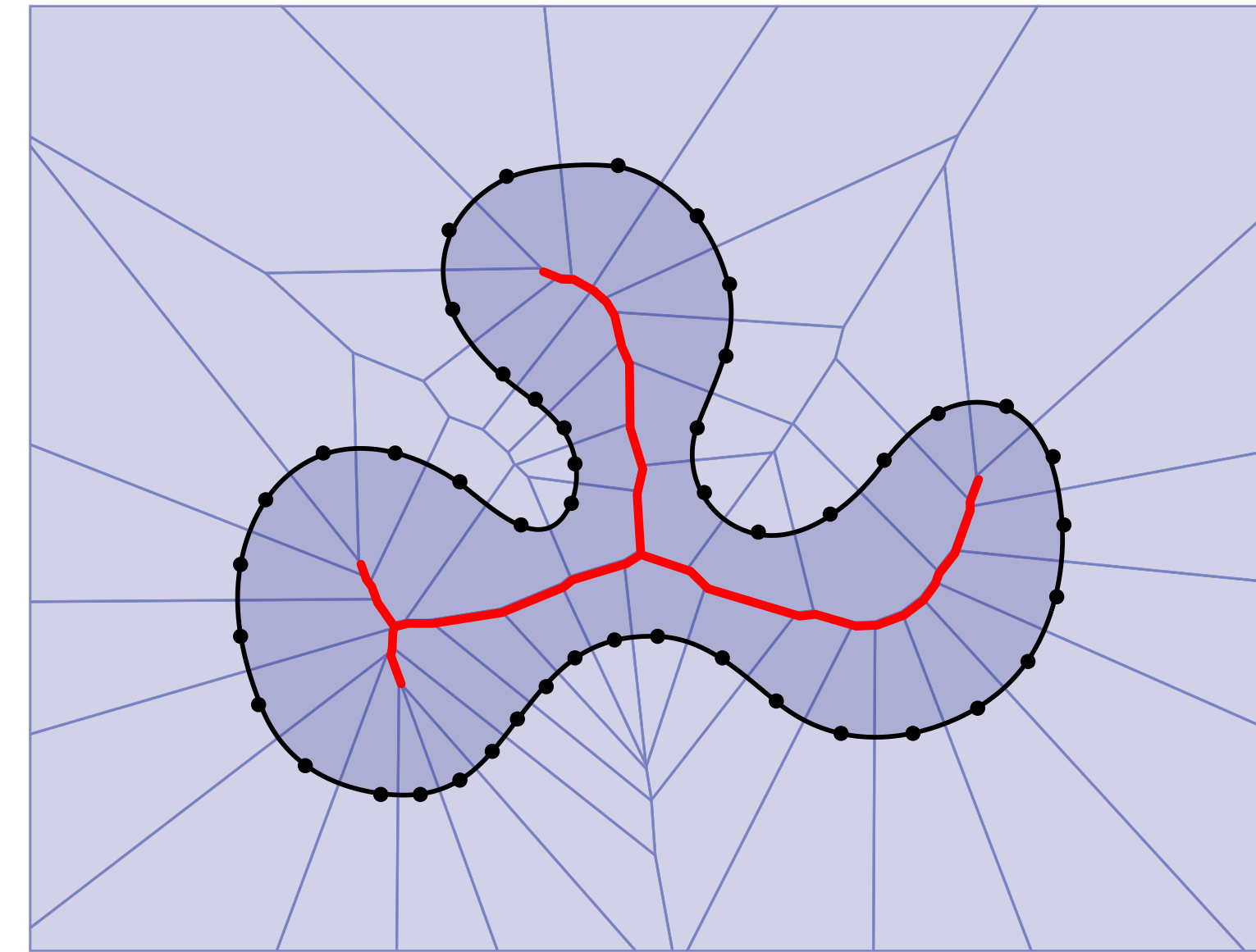
Same definition applies in any dimension—provides notion of “skeleton” for a shape:



Hard to compute exactly (*e.g.*, quadratic pieces); often approximate by simplicial complex.

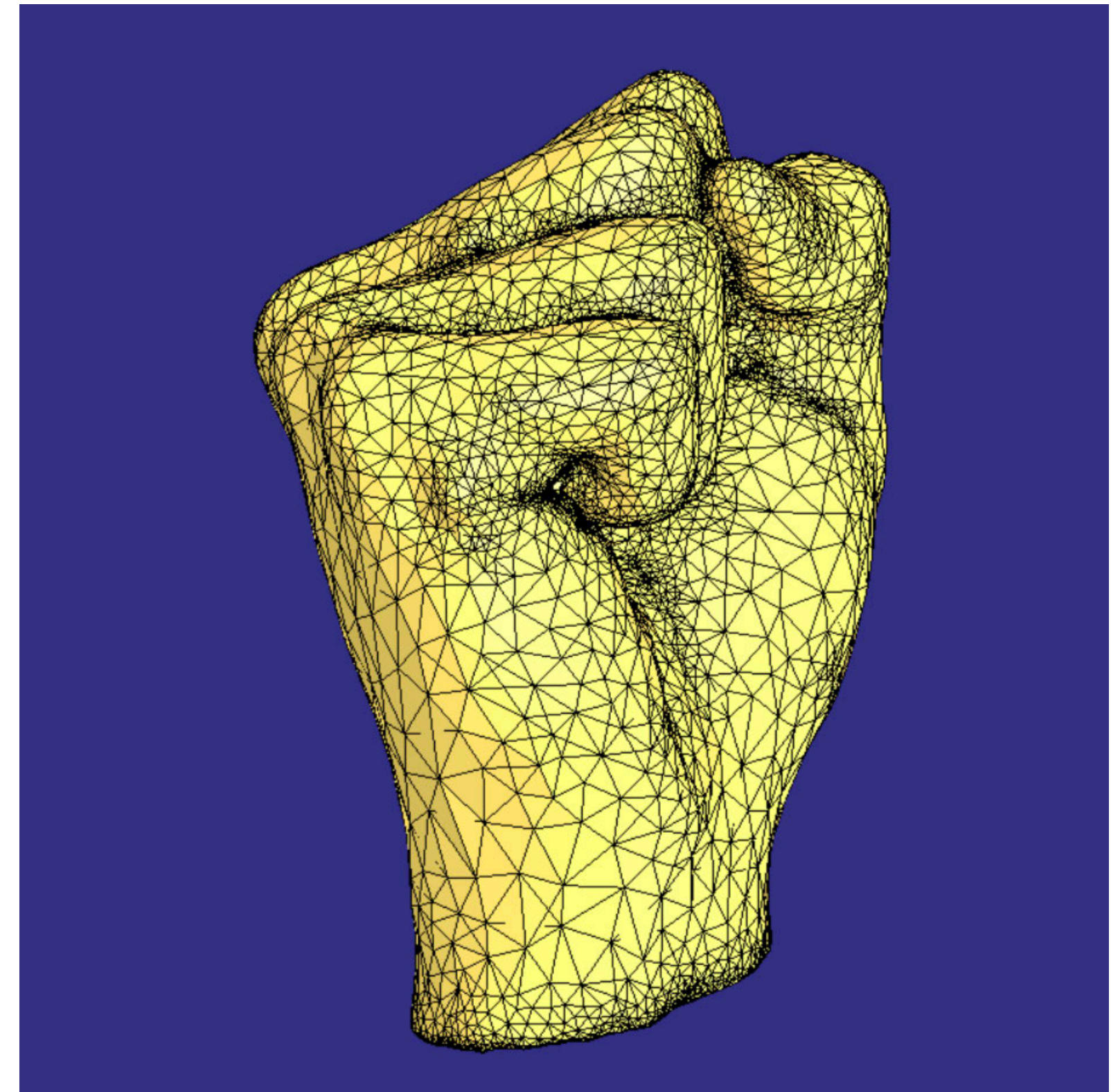
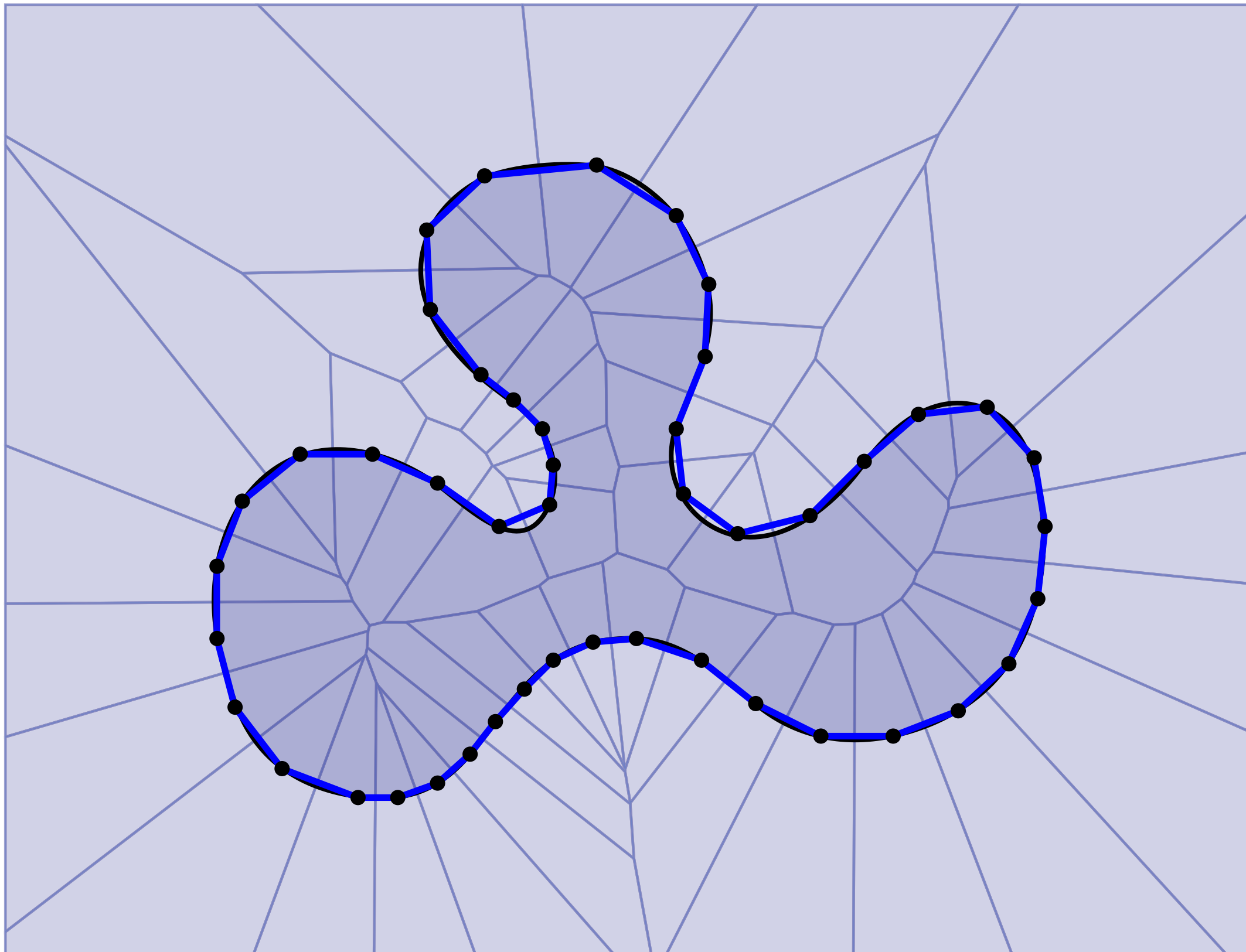
Computing the Medial Axis

- **Many** algorithms for computing / approximating medial axis & other “shape skeletons”
- One line of thought: use *Voronoi diagram* as starting point:
 - sample points on boundary
 - compute Voronoi diagram
 - keep “short” facets of tall / skinny cells
- With enough points, get correct topology



Medial Axis & Surface Reconstruction

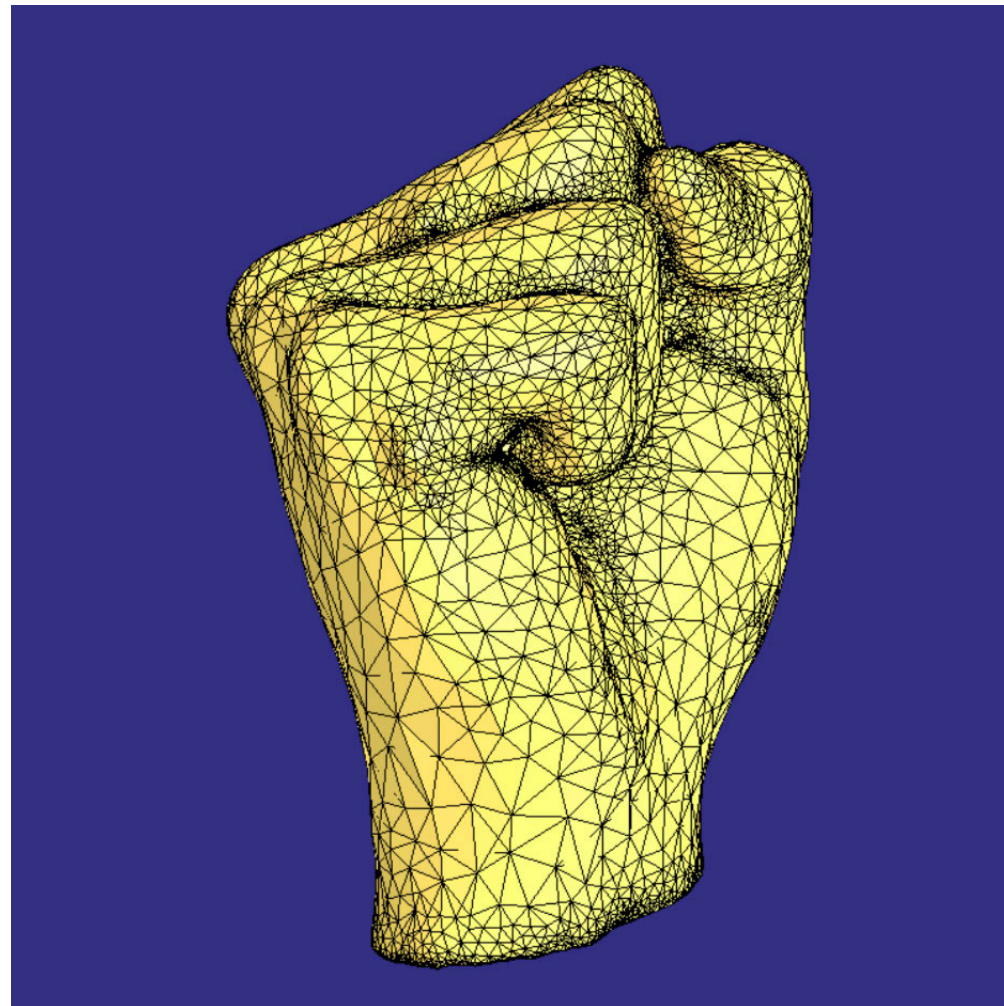
- Can use similar approach for **surface reconstruction** from points
 - connect *centers* of skinny cells that meet along “long” edges
- In 3D, gives surface reconstruction with guarantees on topology (w/ enough points)



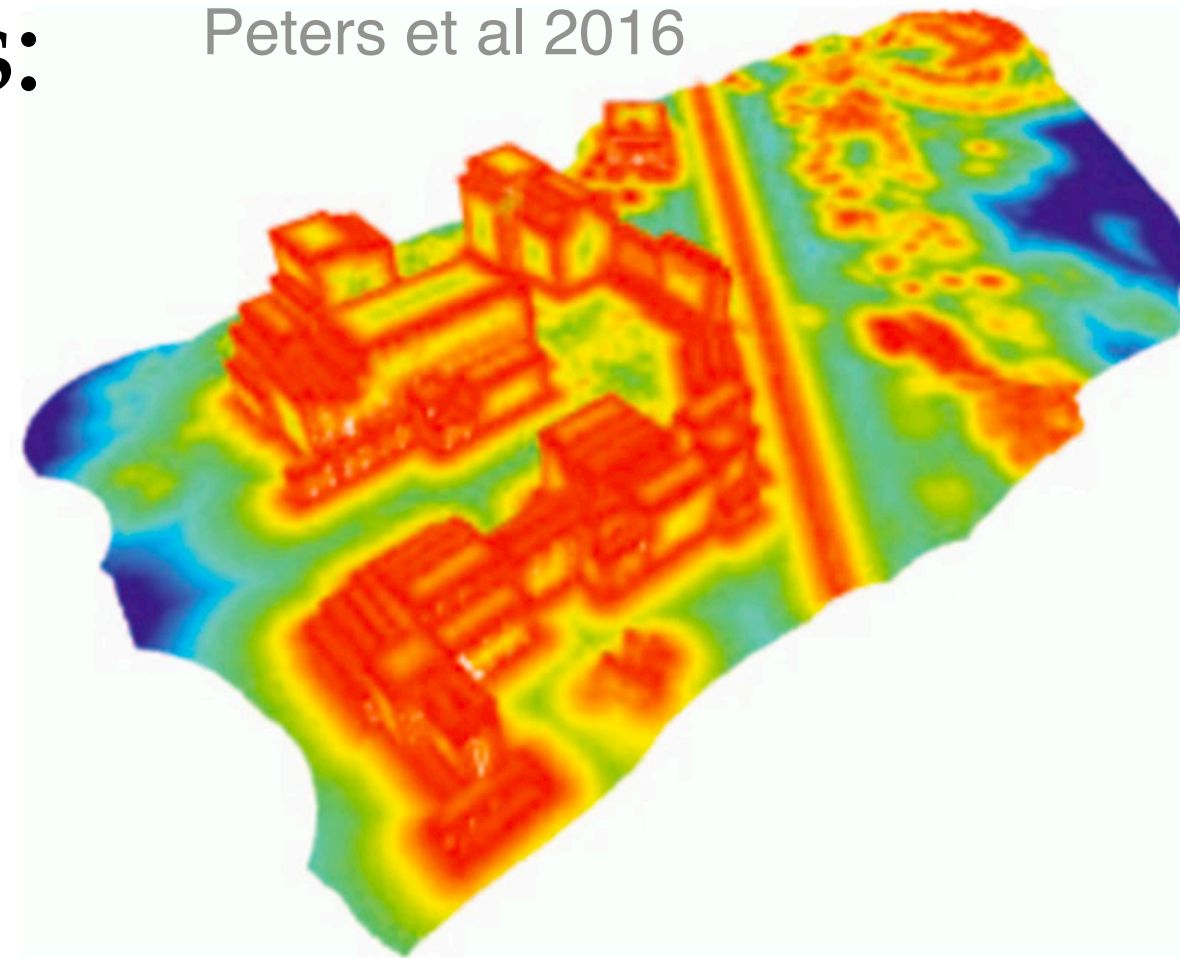
Amenta et al, “A New Voronoi-Based Surface Reconstruction Algorithm”

Medial Axis—Applications

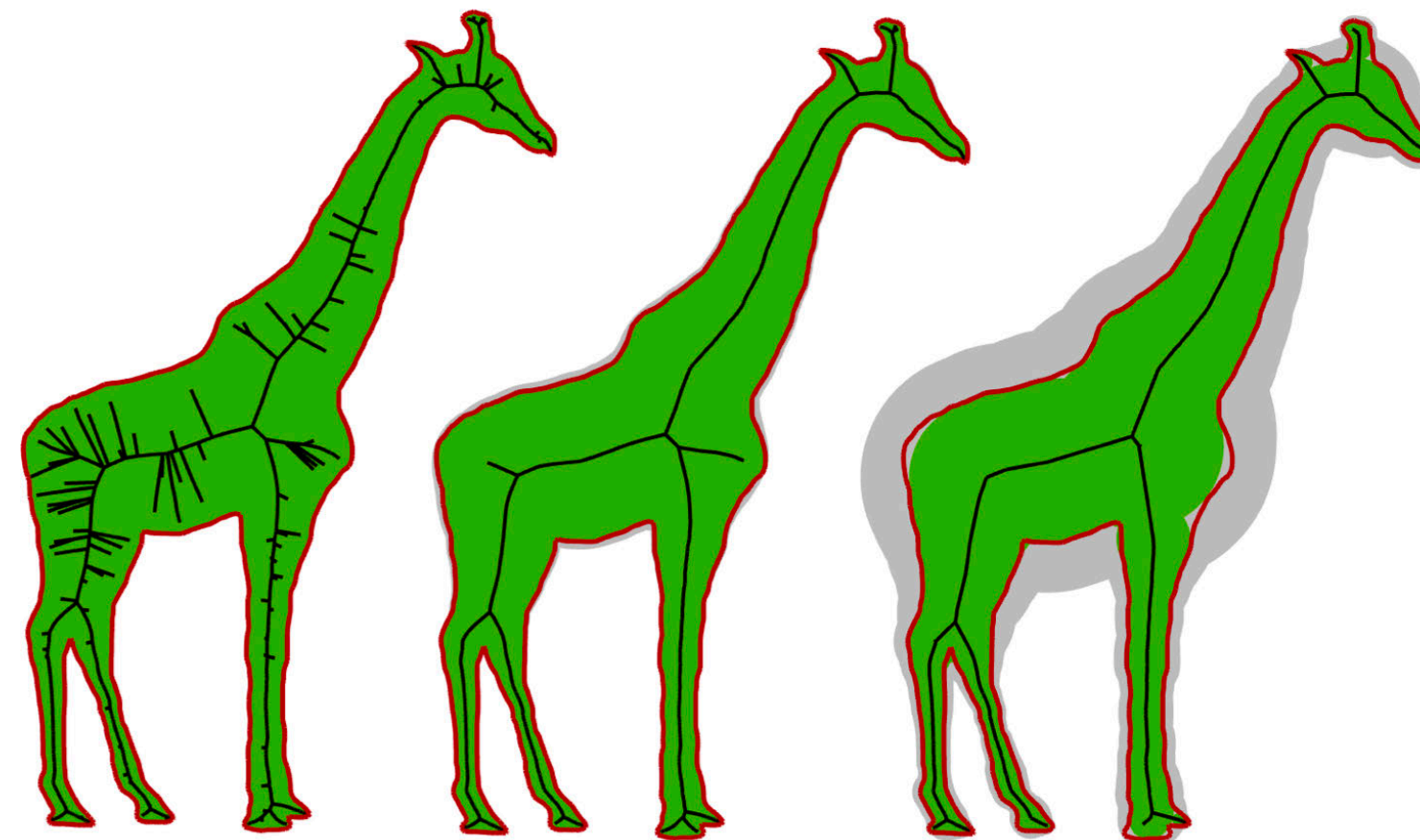
- Many applications of medial axis:
 - surface reconstruction
 - shape skeletons
 - local feature size
 - fast collision detection
 - fluid simulation
 - ...



Amenta et al 1998

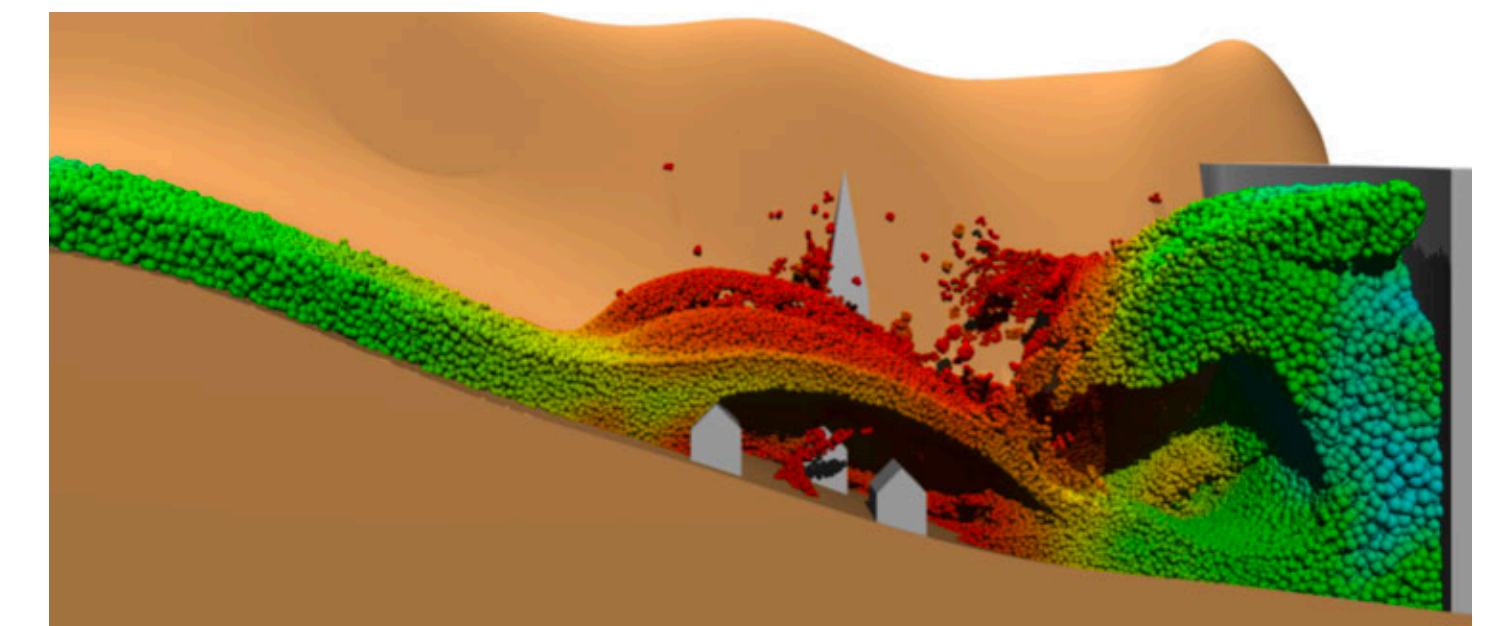
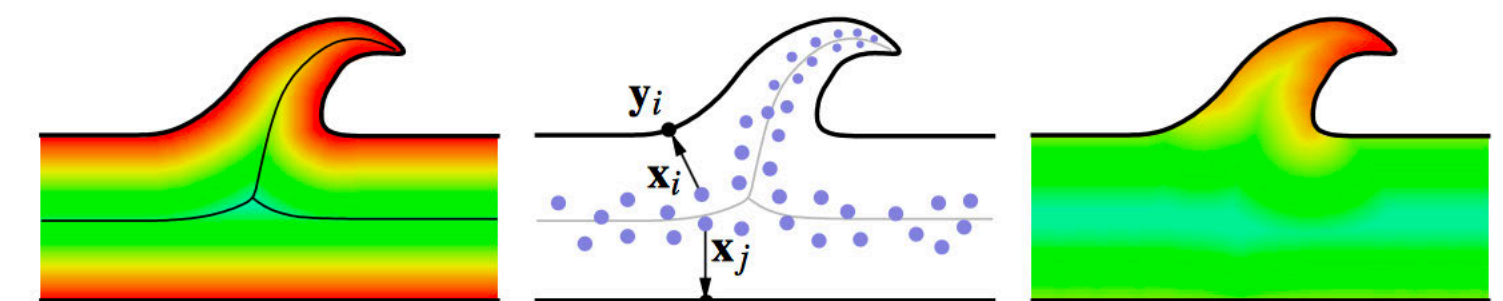
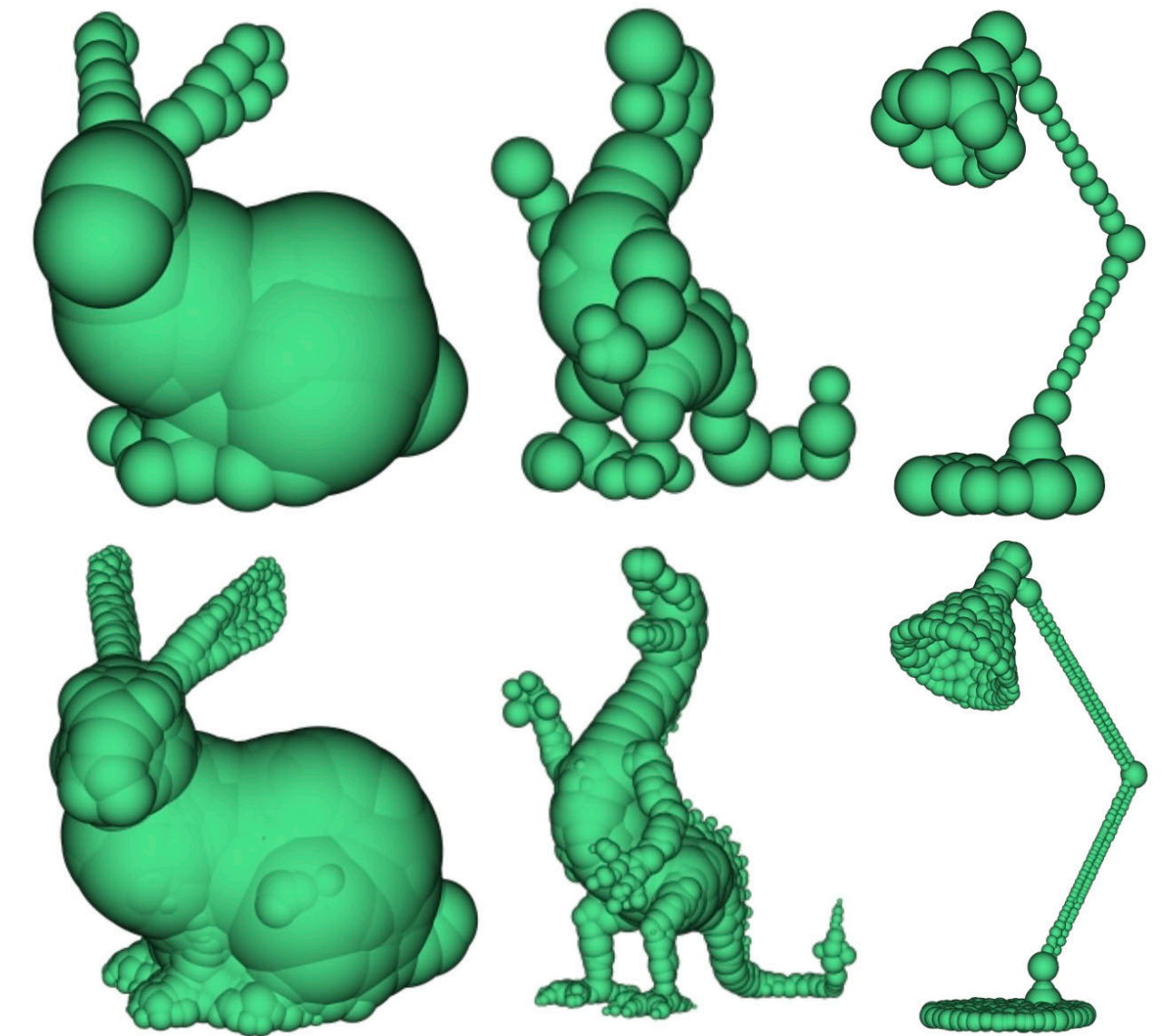


Peters et al 2016

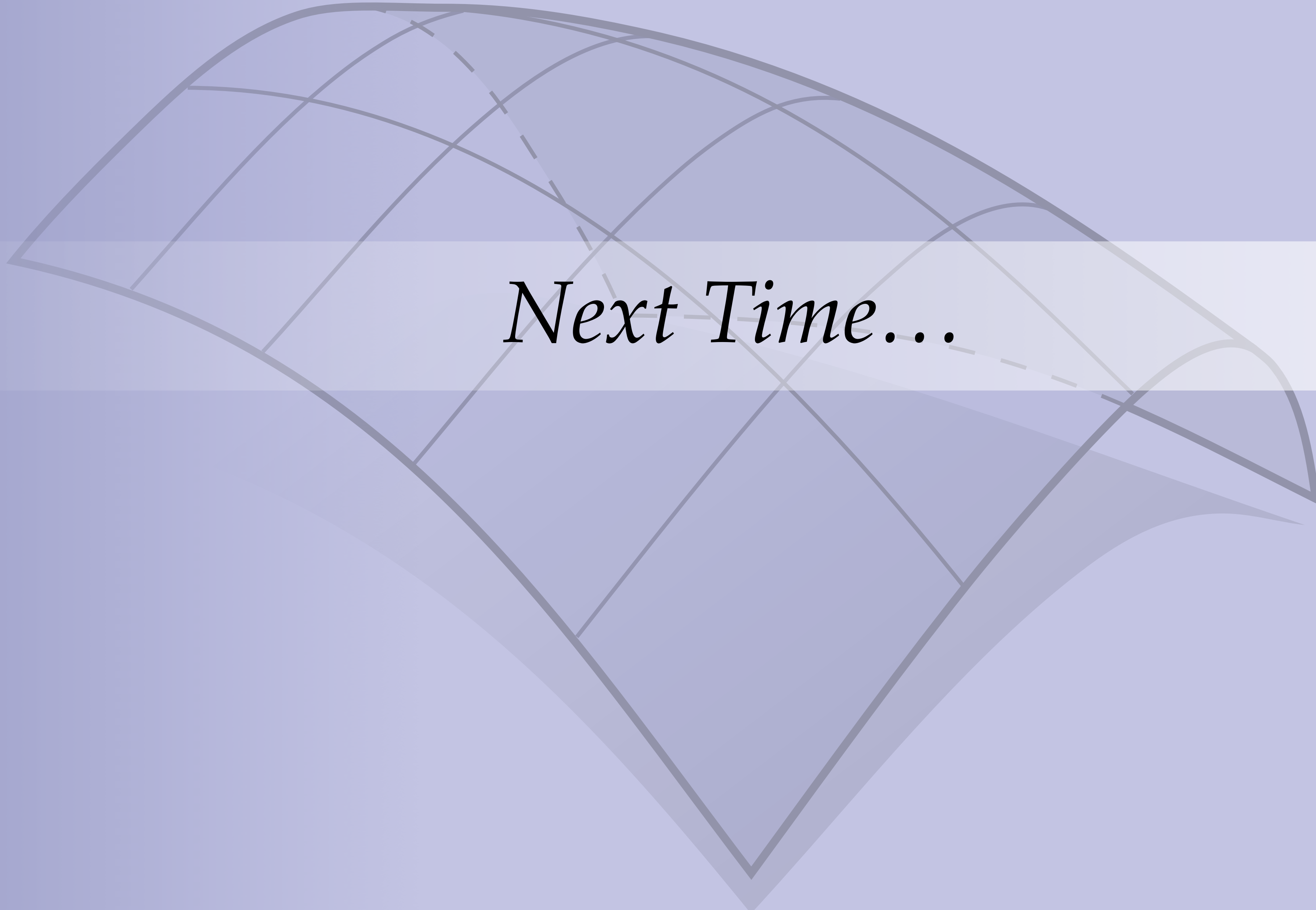


Giesen et al 2009

Bradshaw & Sullivan 2004



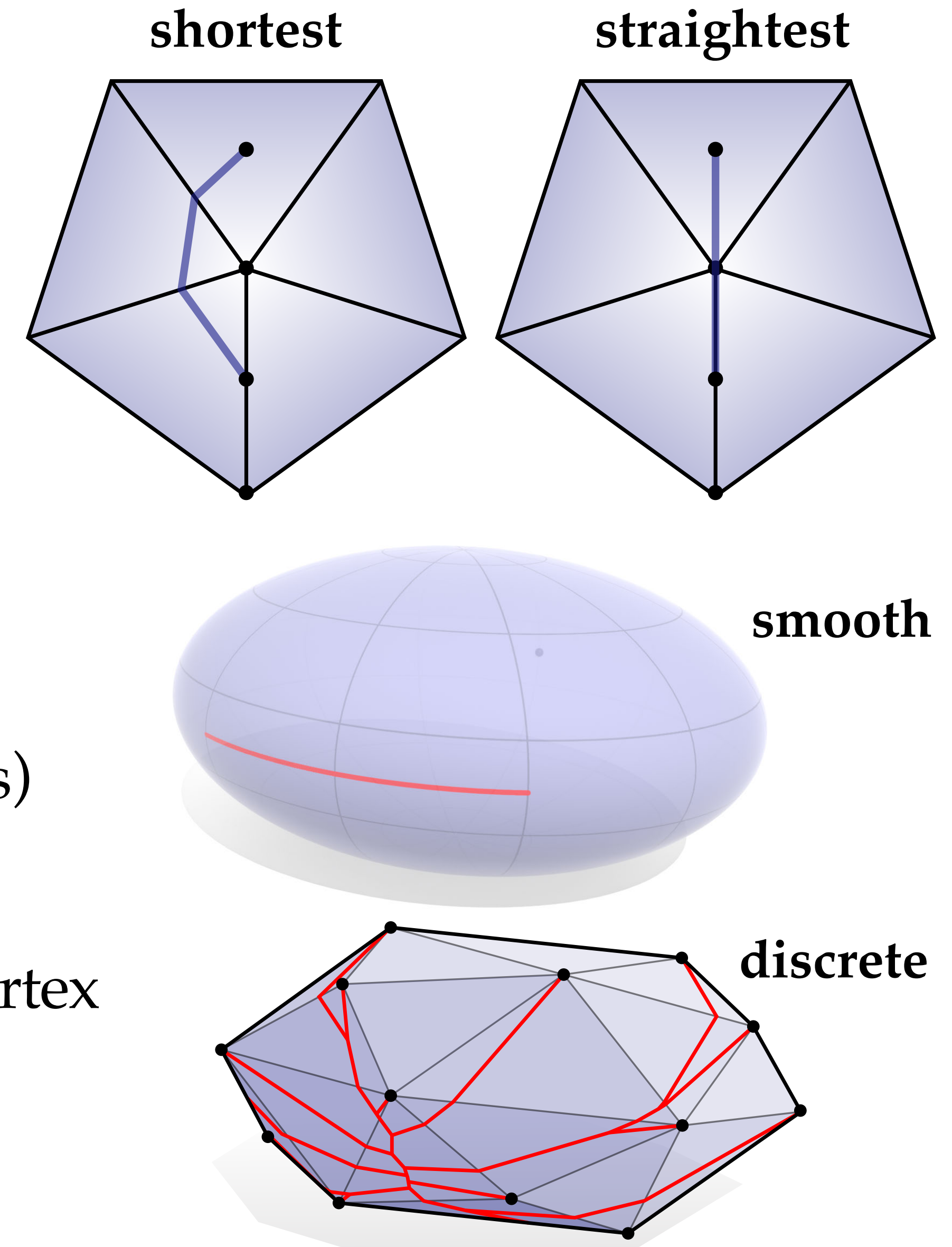
Adams et al 2007



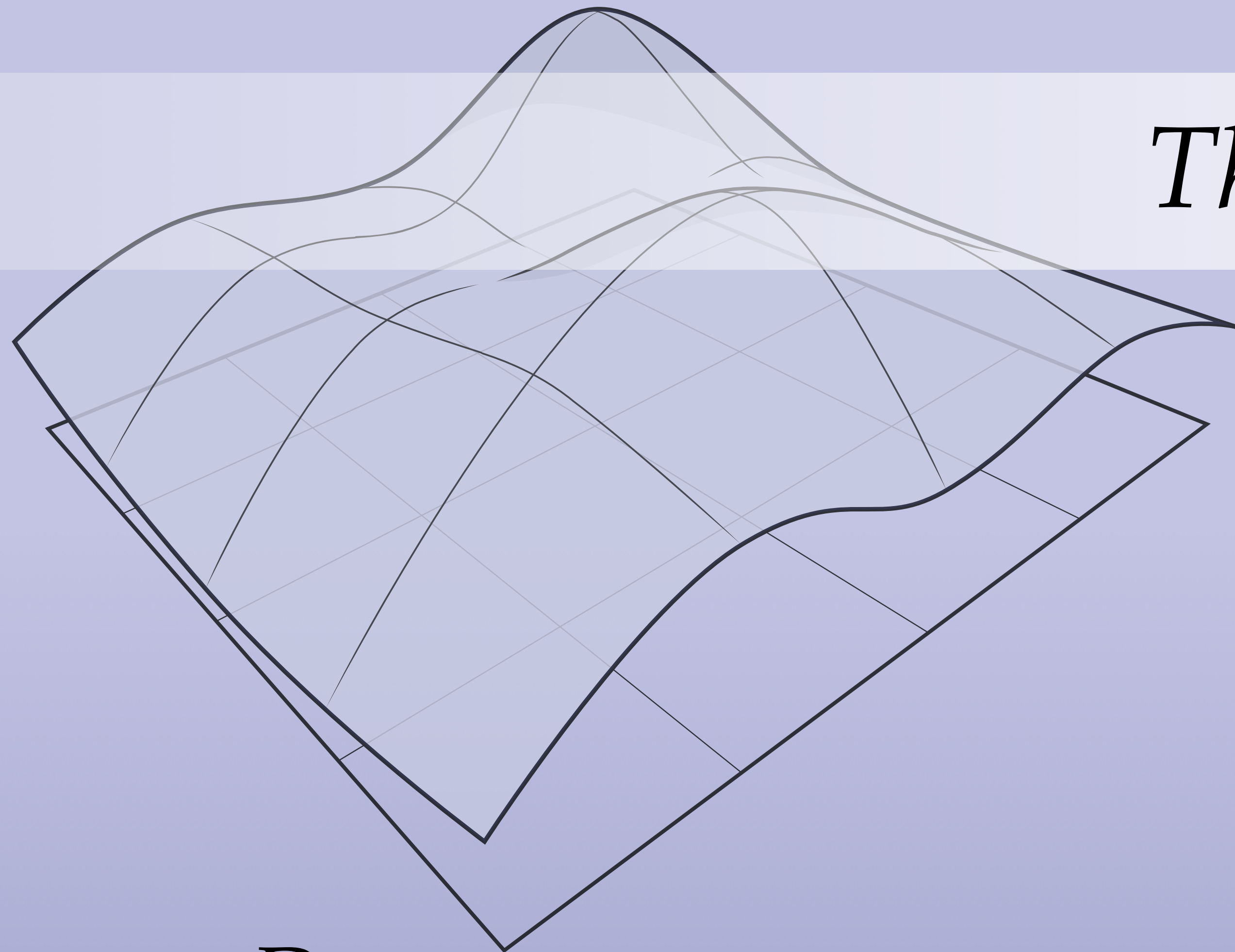
Next Time...

Next Time: Straightest Geodesics

- In smooth setting, several equivalent characterizations:
 - shortest (harmonic)
 - straightest (zero curvature, zero acceleration)
- In discrete setting, characterizations no longer agree!
 - **shortest** natural for boundary value problem
 - **straightest** natural for initial value problem
 - *convex*: shortest paths are straightest (but not vice versa)
 - *nonconvex*: shortest may not even be straightest! (saddles)
- *Neither* definition faithfully captures all smooth behavior:
 - (shortest) cut locus / medial axis touches *every* convex vertex
 - (straightest) exponential map is not surjective
- Use the right tool for the job (*and look for other definitions!*)



Thanks!



DISCRETE DIFFERENTIAL
GEOMETRY:
AN APPLIED INTRODUCTION

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